

Redistribution and Reallocation: Monetary Policy with Heterogeneous Returns*

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Abstract

This paper studies how the aggregate effects of monetary policy depend on wealth inequality. In a model with heterogeneous returns to wealth and endogenous aggregate productivity, a policy rate cut redistributes assets to more productive households, increasing aggregate productivity by reallocating capital to more productive investors. The strength of this redistributive channel is hump-shaped in wealth inequality: it is weak when inequality is either very low or very high and is strongest at intermediate levels. As a result, the overall effect of a change in the policy rate on aggregates is hump-shaped in wealth inequality. When calibrated to the data, the model is able to account for the empirical effects of monetary policy on total factor productivity and inequality. Furthermore, the model implies that the increase in wealth inequality over the past fifty years can help explain the empirically observed decrease in the effect of monetary policy on output.

KEYWORDS: Monetary policy, Wealth distribution, Return heterogeneity, Entrepreneurs, Financial frictions
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1 INTRODUCTION

Over the past fifty years, the effect of monetary policy on economic aggregates has decreased,¹ while wealth inequality has risen. These trends are difficult to reconcile in existing heterogeneous-agent New Keynesian (“HANK”) models: these models, which have emphasized heterogeneity in labor income and marginal propensities to consume, have focused on how household heterogeneity amplifies the effects of monetary policy. However, these models have largely overlooked a critical source of heterogeneity: households in reality earn different returns on their wealth.

Heterogeneity in returns to wealth is a key driver of wealth inequality in the data.² Wealthy households generate high returns by owning assets such as stocks and closely-held businesses, and amplify these returns through leverage, capitalizing on a spread between their private returns and borrowing costs. Expansionary monetary policy lowers borrowing costs and boosts asset values, directly benefiting these wealthy households. This fact is supported in the data: I confirm the empirical finding that wealth inequality rises following a monetary expansion.³ These distributional effects have critical but under-examined implications for aggregate outcomes.

In particular: because wealthy households own productive assets, redistributing wealth towards these households reallocates capital to more productive uses. This intuition is supported in the data: empirically, expansionary monetary policy increases aggregate productivity. An understanding of how monetary policy affects productivity is crucial to the conduct of such policy: productivity alters the tradeoff between output and inflation, which is the central tradeoff that central banks face. Despite this importance, existing HANK models have not accounted for this feature of the data. This gap leaves under-studied the quantitative significance of changes in productivity for the overall effect of monetary policy — as well as how long-term changes to the distribution of wealth may alter how the economy responds to monetary policy.

I address this gap and show that redistribution of assets towards wealthy, high-return households is a key component of the mechanism by which changes to monetary policy influence the economy. I construct a model in which households can operate private businesses, and differ in the productivity with which they do so. Households may operate these businesses using two sources of funding: their own assets, and bonds that they may issue up to a collateral constraint. In equilibrium, my model captures not just wealth and return heterogeneity present in the data, but the correlation between the two: wealthier households earn higher returns. Furthermore, the spread in household returns is endogenous to both the underlying wealth distribution and monetary policy. In this environment, a fall in the nominal interest rate redistributes capital to wealthy, high-return households through two channels. Increases in aggregate demand increase both prices and profit margins; the former devalues high-return households’ debt while the latter increases their real returns.

The redistributive effect of policy in my model formalizes a mechanism present – but not studied – in Kiyotaki (1998). Building on the “Financial Accelerator” literature (Carlstrom and Fuerst, 1997; Bernanke et al., 1999), Kiyotaki (1998) hypothesized that monetary policy shifts assets between with entrepreneurs who

¹See, e.g., Boivin et al., 2010.

²See, for instance, Benhabib et al., 2019, who estimate a quantitative model of the U.S. wealth and income distributions and uncover the central role of return shocks in generating the dispersion in wealth observed in the data.

³See a review in Colciago et al. (2019), or examples in Feilich (2021) and Medlin (2023).

are *ex-ante* heterogeneous, thereby increasing productivity. While the redistributive channel in my model was implicitly present in these papers, until now its relevance for monetary policy has not been uncovered.

These channels of redistribution enable my model to account for the joint increase in wealth inequality and aggregate productivity following monetary expansions in the data — an empirical pattern that existing HANK frameworks have not explained. This literature has emphasized the role of *labor* income risk (McKay et al., 2016; Kaplan et al., 2018; Auclert et al., 2020), focusing on the presence of borrowing-constrained agents with high marginal propensities to consume out of current income. However, these models have assumed that aggregate productivity is exogenous and independent of the wealth distribution. Furthermore, borrowers — who benefit from devaluing of debts — in these models are *poor* households. By incorporating on heterogeneity in returns, and allowing for wealthy households to take on leverage, my model captures features of monetary transmission in the data that these models have missed.

Understanding how productivity responds in a monetary expansion is critical because productivity alters the output-inflation tradeoff, which determines the welfare effects of monetary policy. In existing HANK models, household heterogeneity amplifies policy on the *demand* side: expansions increase the incomes of high-MPC households, resulting in a steeper Phillips curve than in a representative-agent model (Kaplan et al., 2018). In contrast, amplification in my model occurs on the *supply* side, as assets shift to more productive investors. This reallocation results in lower inflation for a given increase in output — a flatter Phillips curve. Redistribution in my baseline calibration flattens the Phillips curve by about 45% relative to a counterfactual with a fixed wealth distribution, implying a meaningfully different policy tradeoff.

As productivity is endogenous to monetary policy, the overall effect of a policy change depends on how much the change alters productivity by redistributing assets. A natural question, then, is how the magnitude of redistribution that accompanies a monetary easing depends on the level of wealth concentration present when the policy change is made. I show that the degree of redistribution that follows a monetary expansion — and thus its effects on productivity and output — are all hump-shaped in the initial degree of wealth inequality. When correctly accounting for the endogeneity of productivity to monetary policy, the monetary transmission is stronger at intermediate values of wealth inequality, and vanishes at very high or very low levels of inequality.

Increases in the concentration of wealth influence the efficacy of monetary policy through two forces, which oppose one another. Intuitively, increases in wealth inequality in this framework imply a tighter link between wealth and returns: the more high-return types can out-accumulate low return types, the greater will be wealth inequality. As such, the higher is wealth inequality when the monetary policy action is taken, the more *targeted* the initial redistribution will be: the households who benefit from the initial redistribution will be those who are perennially highly productive, implying a greater increase in aggregate productivity. A second force, however, countervails as wealth inequality increases. As wealth becomes more concentrated in the hands of high-return households, the return on capital — a wealth-weighted average of individual investors' returns — aligns with the private returns of wealthy households. Following a monetary easing, this alignment of returns results in less redistribution to high-return households: their wealth grows at a similar rate to aggregate wealth, so their share of total wealth changes by less. The reduction in redistribution implies that monetary policy exerts less of an effect on productivity, and thus less of an effect on aggregates. The link

between productivity gains in monetary expansions and the starting level of wealth inequality implies that in order to understand the exact tradeoff that it faces the monetary authority must take wealth concentration into account when formulating policy — even if redistribution of assets is not an explicit goal of the policymaker.

A nascent strand of the HANK literature has studied monetary transmission with heterogeneous firms (Ottonello and Winberry, 2020; Jeenas, 2023), investors (Melcangi and Sterk, 2024), and entrepreneurs (Matusche and Wacks, 2021; González et al., 2024). Across these models, the conclusion is consistent: higher wealth concentration *amplifies* monetary policy’s effect in output and investment. The intuition is that households with high returns have a high propensity to invest out of transitory income shocks, and thus shifting resources to these households increases the economy’s propensity to invest in the aggregate.

This intuition contradicts the data: as wealth inequality has increased since the early 1970s, the measured effects of monetary policy on output have decreased, not increased (Boivin and Giannoni, 2002; Boivin et al., 2010). I show that this intuition in the literature is incomplete: it captures one channel, but misses a second, countervailing channel. My model transparently uncovers both and shows that as wealth inequality increases past threshold level, monetary transmission is *dampened* as the redistribution channel shuts down. Empirically, I find that the second channel has dominated since the 1970s, muting the efficacy of monetary policy and reconciling the observed empirical patterns.

2 MODEL

This section describes the model economy in my paper. I construct a model in which households earn different returns on their investment in equilibrium — returns which are endogenous to both monetary policy and the underlying distribution of wealth itself. To do so, I use a familiar setting of *entrepreneurship* and *financial frictions*: some households have the ability to operate private firms with idiosyncratic productivity, and they may issue debt to fund their operations up to a collateral constraint.

My model is sufficiently parsimonious to admit analytical results while still being able to capture the joint distribution of wealth and returns in the data. The two primary determinants of wealth inequality in the data are the *persistence* and *volatility* in investors’ returns. By varying these underlying parameters, I can transparently expose the channels by which monetary policy redistributes assets across households, and how the strengths of these channels change with underlying inequality.

2.1 INVESTORS’ PROBLEM AND FINANCIAL FRICTIONS

Investor households, indexed by $i \in [0, 1]$, accumulate two assets: capital and risk-free nominal bonds.⁴ Entrepreneurs choose consumption, capital investment, and single-period bonds to maximize their lifetime expected utility:

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \ln(c_{it}) \tag{1}$$

⁴As will become clear shortly, these bonds are risk-free in the sense that their *nominal* rate of return between periods t and $t + 1$ is predetermined in period t . However, their *real* return is subject to risk in the event of unanticipated inflation between these two periods.

Investors split their total income between purchases of consumption goods c_{it} and investment (capital) goods q_{it} ; all goods are identical and hence share a nominal price P_t . Investor households earn income from two sources. These households trade in nominal bonds $D_{i,t}$, which they can issue ($D_{i,t} > 0$) to borrow or purchase ($D_{i,t} < 0$) to lend. Investor households can also operate a private business, which they run by combining capital with labor hired on the spot market to maximize profits in a manner described below. I refer to households operating a business in the current period as *entrepreneurs*, and those saving exclusively in bonds as *lenders*. Denoting entrepreneurs' nominal profit income as $P_t \Upsilon_{i,t}$, investors' budget constraint is

$$P_t (c_{i,t} + q_{i,t}) = P_t \Upsilon_{i,t} + (1 + i_t) D_{i,t} - D_{i,t+1} \quad (2)$$

where P_t is the aggregate price level and i_t the nominal interest rate between $t - 1$ and t , set by the monetary authority at time $t - 1$. Investor i 's capital stock evolves according to $k_{it+1} = (1 - \delta) k_{it} + q_{it}$, where q_{it} is the quantity of investment goods purchased.

Households who operate their private businesses may issue debt to do so, up to a limit. Following Buera and Moll (2015), I assume that households are subject to a collateral constraint of the form

$$D_{it+1} \leq \Lambda P_t k_{it+1}, \quad \Lambda \in [0, 1] \quad (3)$$

This collateral constraint implies that only a proportion θ of the nominal value of the next-period capital stock may be externally financed.

Each entrepreneur household operates a firm that produces homogeneous goods, which are sold to retailers (described below) at nominal price $P_{t,x}$. Entrepreneurs maximize their nominal profit income from running its firm:

$$P_t \Upsilon_{it} = \max_{n_{it}^d} P_{tx} (z_{it} k_{it})^\alpha \left(n_{it}^d \right)^{1-\alpha} - W_t n_{it}^d \quad (4)$$

The following lemma is a well-known result in problems such as (4):

Lemma 1. *Entrepreneurial labor demand is linear in capital, and given by:*

$$n_{it}^d = \left(\frac{1 - \alpha}{W_t / P_t} \frac{P_{tx}}{P_t} \right)^{\frac{1}{\alpha}} z_{it} k_{it} \quad (5)$$

Lemma 1 results from the fact that the problem in (4) is static: entrepreneurs hire labor on the spot market to maximize their profit given their state (z_{it}, k_{it}) and the wage and price W_t and P_{tx} , which they take as given. Defining

$$\omega_t \equiv \alpha \left(\frac{1 - \alpha}{w_t} \right)^{\frac{1-\alpha}{\alpha}} p_{tx}^{\frac{1}{\alpha}} \quad (6)$$

where w_t is the real wage and p_{tx} the real entrepreneurs' price, the budget constraint can be written as

$$P_t c_{it} + P_t k_{it+1} = P_t [\omega_t z_{it} + (1 - \delta)] k_{it} - (1 + i_t) D_{it} + D_{it+1} \quad (7)$$

It is useful to write the entrepreneurs' budget constraint in real terms. To do so, define real bond issuance

as $d_{i,t+1} \equiv D_{i,t+1}/P_t$. With this definition, the budget constraint for entrepreneurial household i in real terms is

$$c_{it} + k_{it+1} = [\omega_t z_{it} + 1 - \delta] k_{it} - (1 + r_t) d_{it} + d_{it+1} \quad (8)$$

Here, r_t is the time- t ex-post real interest rate, defined by the Fisher equation:

$$1 + r_t = (1 + i_t) \frac{P_{t-1}}{P_t} = \frac{1 + i_t}{1 + \pi_t} \quad (9)$$

Note that this interest rate depends on the realized inflation rate π_t . Defining the real net worth as $a_{i,t} \equiv k_{i,t} - d_{i,t}$ I can write the borrowing constraint in real terms as $d_{t+1} \leq \Lambda k_{t+1}$, or substituting the definition of net worth a_t ,

$$k_{t+1} \leq \lambda a_{t+1}, \quad \lambda \equiv \frac{1}{1 - \Lambda} \quad (10)$$

The timing in the economy works as follows. Following production in period t , investor households realize their next-period productivity $z_{i,t+1}$, and choose their next-period capital stock $k_{i,t+1}$ and bond holdings $d_{i,t+1}$ — equivalently, whether to be entrepreneurs or lenders. Because returns are linear in wealth, the choice to be an entrepreneur only depends on idiosyncratic productivity $z_{i,t}$. Furthermore, households who do choose to be entrepreneurs will choose to invest as much as possible:

Lemma 2. *Entrepreneurs' capital and bond choices are corner solutions:*

$$k_{t+1} = \begin{cases} \lambda a_{t+1} & z_{t+1} \geq \underline{z}_{t+1} \\ 0 & z_{t+1} < \underline{z}_{t+1} \end{cases} \quad d_{t+1} = \begin{cases} (\lambda - 1) a_{t+1} & z_{t+1} \geq \underline{z}_{t+1} \\ -a_{t+1} & z_{t+1} < \underline{z}_{t+1} \end{cases} \quad (11)$$

where the cutoff $\underline{z}_{t+1}$, which solves $\omega_{t+1} \underline{z}_{t+1} = \mathbb{E}_t r_{t+1} + \delta$, is the productivity at which an investor is indifferent between private production and public bonds, given expectation of the one period-ahead real interest rate.

Lemma 2 shows that investors are divided into two groups: those above the productivity threshold $\underline{z}_t$, who are entrepreneurs, and those below it, who are lenders. Active entrepreneurs, who earn excess returns on their investment above the risk-free rate, borrow up to their limit and are thus bound by the collateral constraint. Lenders save at the risk-free rate by lending to active entrepreneurs. Due to the linearity of the production technology, the cutoff productivity $\underline{z}_t$ is independent of wealth; instead, $\underline{z}_t$ is a linear function of the risk-free rate r_t and ω_t , which can be thought of as the private return per effective unit of capital zk .

Finally, the assumption of log utility among investor households implies that these households will choose to save a constant fraction β of their financial wealth, post-returns: $a_{i,t+1} = \beta R_t(z) a_{i,t}$. I refer to the object $\beta R_t(z)$ as the marginal propensity to save for type z , $MPS_t(z)$. This object captures the propensity of a household of type z to accumulate out of an increase in *pre-return* wealth. As such, the distribution of $MPS(z)$ will be crucial in determining how redistribution across households translates into aggregate outcomes.

2.2 NOMINAL RIGIDITIES

To introduce nominal rigidities while maintaining tractability, I follow Bernanke et al. (1999) in assuming a three-tiered production structure. Entrepreneurs produce a homogenous good x_t , which is then sold to retailers. Retailers, a continuum of whom are indexed by $j \in [0, 1]$, in turn costlessly differentiate these goods. Retailers sell their output y_{tj} to a final good producer, who aggregates them using a CES technology:

$$Y_t = \left[\int_0^1 y_{tj}^{\frac{\varepsilon-1}{\varepsilon}} dj \right]^{\frac{\varepsilon}{\varepsilon-1}} \quad (12)$$

This assumption on the structure of production allows me to introduce price stickiness in a way that preserves the tractability of the entrepreneurs' problem. It is analytically convenient to assume that entrepreneurs are price takers; otherwise, their investment and savings choices would be intermingled with a forward-looking pricing problem, which would complicate the model without providing any additional insights.

Optimal behavior by the final good aggregator in (12) implies that the demand for variety j is

$$y_{t,j} = \left(\frac{P_{t,j}}{P_t} \right)^{-\varepsilon} Y_t \quad (13)$$

where $P_t \equiv \left(\int P_{t,j}^{1-\varepsilon} dj \right)^{\frac{1}{1-\varepsilon}}$ is the overall price level. Retailer j 's output is then simply $y_{tj} = x_{tj}$, and their marginal cost is $m_t = p_{tx}$. In addition, retailers incur Rotemberg (1982)-style quadratic adjustment costs to change their price:

$$\Theta(P_{tj}, P_{t-1,j}) = \frac{\theta}{2} \left(\frac{P_{tj}}{P_{t-1,j}} - 1 \right)^2 Y_t \quad (14)$$

The retailer incurs an adjustment cost when it wants to update its price relative to its own lagged price. Defining inflation as $\pi_t \equiv \frac{P_t}{P_{t-1}} - 1$, the following lemma describes optimal price-setting behavior by retailers:

Lemma 3. *To first order, inflation satisfies the New Keynesian Phillips Curve, which arises under optimal behavior by retailers:*

$$\pi_t = \frac{\varepsilon}{\theta} (p_{tx} - m^*) + \beta \mathbb{E}_t \pi_{t+1} \quad (15)$$

Here, $m^* = \varepsilon / (\varepsilon - 1)$ is the inverse of the optimal markup in the absence of price rigidities.

The intuition in Lemma 3 is standard. In the presence of price stickiness, retailers raise their prices when they believe that future marginal costs will exceed their long-run optimal level—equivalently, retailers raise current prices when they believe that, in the future, markups will fall *below* their long-run optimal level.

2.3 EQUILIBRIUM

There are two actors needed to close the model: workers, and a monetary authority. I assume that workers—who supply their labor to entrepreneurs at real wage w_t —cannot borrow or save, and are thus constrained to be hand-to-mouth. Workers are, however, free to adjust their labor supply in response to movements in the

real wage. Worker households are all identical, and have preferences as in Greenwood et al. (1988):

$$U^w(C_t^w, N_t^w) = \frac{1}{1-\gamma} \left(C_t^w - \frac{(N_t^w)^{1+\eta}}{1+\eta} \right)^{1-\gamma} \quad (16)$$

In addition, worker households own the retailers, and receive the profits of these firms as real dividends T_t . Workers' budget constraint in real terms is simply $C_t = w_t N_t + T_t$. The labor supplied by the households is given by

The monetary authority sets the nominal interest rate i_t according to a Taylor rule:

$$i_{t+1} = \bar{r} + \phi_\pi \pi_t + \nu_t \quad (17)$$

Recall that i_t dictates the nominal cost that an entrepreneur pays for outside financing. In order to ensure notational consistency, I date all interest rates according to when they are earned, rather than when they are set. As such, the nominal rate i_{t+1} in (17) is set at time t , and dictates the interest rate on nominal debt issued in period t and maturing in period $t+1$. The term ν_t is an exogenous, stochastic innovation; I will use this shock to measure the impact of monetary policy in my model.

With these remaining actors thus established, I can define an equilibrium in my model economy:

Definition 1. An equilibrium is a sequence of prices $\{P_t, P_{tx}, W_t\}$, aggregates $\{C_t^e, C_t^w, N_t, Y_t, K_t, Z_t\}$, interest rates $\{i_t, r_t\}$, a path for inflation π_t , a sequence of aggregate shocks ν_t , and a sequence of distributions $\{g_t(a, z)\}$ over the idiosyncratic states for entrepreneurs such that:

1. Entrepreneurs, workers, retailers, and the final good producer all maximize their respective objectives,
2. The monetary authority sets the nominal interest rate in accordance with the Taylor rule in (17), given an exogenous sequence for the shock ν_t ,
3. Prices clear markets:

$$K_t = \int_0^{\bar{z}} \int_0^\infty a g_t(a, z) \, da \, dz \quad (18)$$

$$N_t^w = N_t^d \quad (19)$$

$$C_t^e + C_t^w + K_{t+1} + \Theta(\pi_t) = Y_t + (1 - \delta) K_t \quad (20)$$

To track the equilibrium distribution of wealth, I use *wealth shares*:

$$s_t(z) \equiv \frac{1}{K_t} \int_0^\infty a g_t(a, z) \, da \quad (21)$$

As in Moll (2014), among others, the wealth share $s_t(z)$ denotes the share of aggregate wealth held by agents of type z . There are a number of reasons why these objects are a convenient tool for studying the behavior of the model. First, the shares $s_t(z)$ act as a density: they are nonnegative for all z , and integrate to one:

$\int_0^{\bar{z}} s_t(z) dz = 1$ for all t . As such, I can define the analogous cumulative share:

$$S_t(z) \equiv \int_0^z s_t(x) dx \quad (22)$$

Second, note that because returns are linear in wealth, individual wealth follows a random growth process.⁵ As a result, the joint distribution $g_t(a, z)$ does *not* admit a stationary measure: the log of individual wealth a_t follows a random walk, and thus the cross-sectional variance of a_t grows without bound in t . However, it can be demonstrated that the wealth shares $s_t(z)$ *do* admit a stationary measure. This result allows me to study the evolution of the wealth distribution without needing to augment the model with an assumption to deliver a stationary measure over wealth, such as random death and annuity markets (as in Gouin-Bonenfant and Toda, 2019) or a hard borrowing limit.

Using the definition of wealth shares in (21), aggregate quantities are as follows:

Proposition 1. *Aggregate quantities satisfy*

$$Y_t = (Z_t K_t)^\alpha N_t^{1-\alpha} \quad (23)$$

$$K_{t+1} = \beta \{ \alpha p_{tx} Y_t + (1 - \delta) K_t \} \quad (24)$$

Aggregate productivity is a function of the wealth distribution $s_t(z)$:

$$Z_t = \mathbb{E}_{s_t} [z | z > \underline{z}_t] \quad (25)$$

Given a path for wealth shares $s_t(z)$, the cutoff productivity $\underline{z}_t$ is pinned down by capital market clearing:

$$1 = \lambda (1 - S_t(\underline{z}_t)) \quad (26)$$

Factor prices are

$$w_t = (1 - \alpha) p_{tx} \left(\frac{Z_t K_t}{N_t} \right)^\alpha \quad (27)$$

$$\mathbb{E}_{t-1} r_t = \alpha p_{tx} Z_t^\alpha \left(\frac{N_t}{K_t} \right)^{1-\alpha} \frac{\underline{z}_t}{Z_t} - \delta \quad (28)$$

Returns are given by

$$\omega_t = \alpha p_{tx} \left(\frac{N_t}{Z_t K_t} \right)^{1-\alpha} \quad (29)$$

$$R_{tK} = 1 + \omega_t Z_t - \delta \quad (30)$$

⁵See Gabaix (2009) for a study of random growth processes in economics, and Benhabib et al. (2015) for an example of how this process gives rise to wealth distributions in models that share the “fat-tailed” (Pareto) nature of their empirical counterparts.

The return to an entrepreneur of type z is given by

$$R_t(z) = 1 + i_{t-1} - \pi_t + \lambda \cdot \max \{0, \omega_t z - (i_{t-1} - \pi_t + \delta)\} \quad (31)$$

The proof of Proposition 1 can be found in Appendix A.1. This proposition is an analogue of similar results in Moll (2014) and Buera and Moll (2015) adapted to an economy with nominal rigidities, elastic labor supply, and monopolistic competition in the goods market. Equations (23) and (24) show that this economy behaves as one with a representative firm, with a key difference: aggregate TFP is an endogenous result of the wealth distribution among investor households, per equation (25). In equation (27), the real wage is given by the real value of the aggregate marginal product of labor hired by entrepreneurs. The same is generally *not* true for the ex-ante expected risk-free real rate $\mathbb{E}r_t$: in equation (28), this object is equal to the aggregate marginal return to capital, weighted by $\underline{z}_t/Z_t$.⁶ In this economy, capital market frictions prevent the return from investment to be equated with outside savings. As a corollary, the two are equated in the case that $\underline{z}_t = Z_t$, which is the case when capital markets are frictionless, $\lambda = \infty$. Note as well that both factor prices, as well as the returns R_{tK} and ω_t , move with the price paid to entrepreneurs by retailers for their goods.

In equation (30), the aggregate return to capital is derived from

$$\begin{aligned} R_{tK} &= \int_0^{\bar{z}} R_t(z) s_t(z) dz \\ &= \mathbb{E}_{s_t} [R_t(z)] \end{aligned}$$

The aggregate return on capital is the average return across all entrepreneurs, weighted by their respective wealth shares. Finally, entrepreneurs' returns, per equation (31), exhibit a few key properties that will later drive my results. First, entrepreneurs with $z > \underline{z}_t$ are able to earn excess returns above the ex-ante risk-free rate $i_{t-1} - \pi_t$, due to their ability to make leveraged investments into their inside firm. In partial equilibrium, equation (31) previews the differential impact of a change in real rates on entrepreneurs of different productivity. Returns can also be written as

$$R_t(z) = \begin{cases} 1 + i_{t-1} - \pi_t & z \leq \underline{z}_t \\ 1 - (\lambda - 1)(i_{t-1} - \pi_t) + \lambda(\omega_t z - \delta) & z > \underline{z}_t \end{cases} \quad (32)$$

From (32) it is immediately obvious that a fall in the real rate $i_{t-1} - \pi_t$ lowers the returns of savers, who *earn* this rate, and raises that of active entrepreneurs, who *pay* this rate to borrow capital. Additionally, the expression of returns in (32) makes clear the role of inflation in driving redistribution in this model: an unexpected increase in π_t reduces nominal obligations, redistributing from low types (lenders) to high types (borrowers). Crucially, π_t is realized one period *after* $\underline{z}_t$ has been determined: entrepreneurs operate their firms at time t with capital installed at time $t - 1$, and thus agents cannot switch from being inactive to active

⁶The expectations operator indicates that the ex-post real rate is subject to inflation risk. In the absence of nominal rigidities, Equation (28) would always hold. In order to study the redistributive effects of unanticipated inflation, I leave open the possibility that the ex-post real rates may differ from their ex-ante expectations. In the event that inflation is not equal to its ex-ante expectation, this equation will hold for the expected risk-free rate, upon which time- t contracts are based, but *not* for the ex-post rate r_{t+1} .

following inflation that deviates from expectations.

Before studying the properties of this equilibrium, I emphasize its generality in understanding the role of return heterogeneity in transmitting monetary policy. From the perspective of policy, the key features of this model are that (i) agents earn heterogeneous returns and (ii) take on leverage, (iii) the aggregate return to capital is a wealth-weighted average of returns above a cutoff for participation in the risky asset, and (iii) this cutoff is endogenously determined so as to clear the capital market. The most restrictive assumption that I make is that these firms are unable to issue equity, and thus must fund all investments above their existing wealth with debt. Nevertheless, these forces are likely to be present even when households have risky returns resulting from ownership of *public* firms as well.

2.4 DISTRIBUTION OF RETURNS

To study analytically the effects of a monetary shock in my model economy, I make the following assumption on individual entrepreneurs' productivity:

Assumption 1. *With probability p , an entrepreneur will maintain his productivity from one period to the next, $z_{t+1} = z_t$. With probability $1 - p$, meanwhile, he draws his next-period productivity at random from the time-invariant distribution $F(z)$. Furthermore, I assume $F(z)$ is lognormal, and parameterized by σ_z :*

$$\ln z \sim \mathcal{N}(-\sigma_z^2/2, \sigma_z^2)$$

Assumption 1 allows for gains in tractability while maintaining rich heterogeneity in wealth and returns. With this assumption, the underlying mass of households in the economy will be given by $F(z)$, and p is the autocorrelation of z_t and z_{t+1} . Persistence in returns is the empirically relevant case: autocorrelated returns have been documented in the micro data (see, e.g., DeBacker et al., 2023). Furthermore, persistence in returns allows the model to capture the empirical correlation between returns and wealth (Xavier, 2021): if investors keep their returns over time, then high-productivity entrepreneurs can accumulate savings out of perennial high returns to ease the financial constraint in (10). In the long run, then, persistent returns give rise to an equilibrium in which wealthier households earn higher returns — as in the data.

Under assumption 1, wealth shares $s_t(z)$ defined in (21) evolve as follows:

Proposition 2. *The wealth share of type z , $s_t(z)$, evolves according to*

$$s_{t+1}(z) = p \frac{R_t(z)}{R_{tK}} s_t(z) + (1 - p) f(z) \quad (33)$$

where R_{tK} is the aggregate (wealth-share weighted average) return to capital, as defined in Proposition 1.

For the proof of Proposition 2, see Appendix A.2. The law of motion in equation (33) has an intuitive interpretation. There are two sources of change in the wealth share $s_{t+1}(z)$: entrepreneurs who retain their type ($z_{t+1} = z_t$), and entrepreneurs who transition to type z at $t + 1$ from some other type ($z_t \neq z_{t+1}$). For each source of change, the sign of its contribution (whether it increases or decreases $s_{t+1}(z)$ relative to $s_t(z)$) depends on the returns of the agents in question, relative to the aggregate return on capital. For agents who

retain their type: if the time- t return $R_t(z)$ is greater than the aggregate return to capital, then the wealth of agents of type z grows faster than the overall capital stock, and their share increases. Agents transitioning to type z from some other z' – the $(1-p)f(z)$ term in (33) – on average earn the aggregate return on capital R_{tK} by definition, hence the coefficient of one on this term.

2.5 LONG-RUN WEALTH INEQUALITY

One of the primary questions of my paper is how the distribution of wealth at the time of a policy shock influences the economy's aggregate response to that shock. Here, I analyze properties of the long-run steady state equilibrium that will be critical to answering this question. I assume that prior to the unexpected change in monetary policy ν_0 , the economy is in its *long-run, zero-inflation steady state*. Equation (33) implies that in the steady state, wealth shares are given by

$$s(z) = \frac{1-p}{1-\beta p R(z)} f(z) \quad (34)$$

Equation (34) uses the fact that in the steady state, the return to capital is pinned down by the entrepreneurs' discount factor:

$$R_K = 1 - \omega z - \delta = \frac{1}{\beta} \quad (35)$$

where, per Equation (29), the steady-state return per effective unit of capital zk is

$$\omega = \alpha p_x \left(\frac{N}{ZK} \right)^{1-\alpha}$$

. The wealth shares $s(z)$ are a mixture of the marginal distribution $f(z)$, which determines the mass of agents of type z , and the steady-state returns earned by type z , given by

$$R(z) = 1 + \omega z_\lambda(z) - \delta \quad (36)$$

where

$$z_\lambda(z) \equiv \begin{cases} z + (\lambda - 1)(z - \underline{z}) & z > \underline{z} \\ \underline{z} & z \leq \underline{z} \end{cases} \quad (37)$$

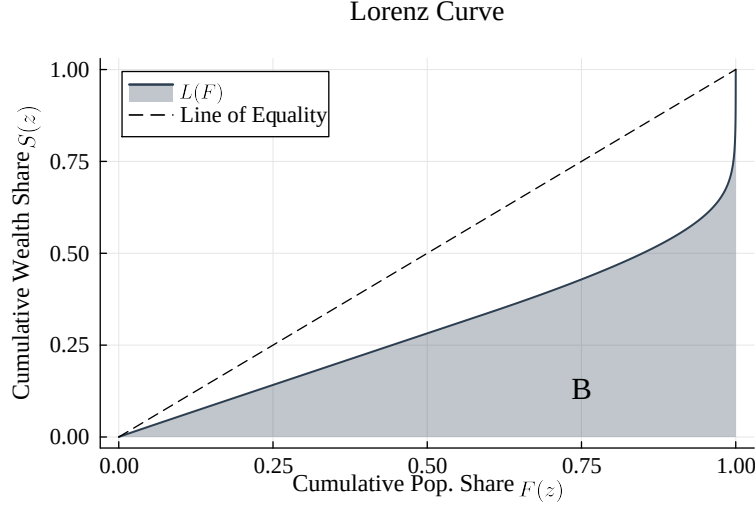
is the effective return to an entrepreneur of type z , taking into account their choice between investing and lending, and leverage. The real interest rate is the effective productivity of the *marginal* entrepreneur:

$$r = \omega \underline{z} - \delta \quad (38)$$

The price for entrepreneurial goods is equal to its optimal level in the absence of nominal rigidities:

$$p_x = 1/\mathcal{M}^* = \frac{\varepsilon - 1}{\varepsilon} \quad (39)$$

Figure 1: Lorenz Curve



Aggregate productivity is determined by the allocation of wealth among entrepreneurs, as described by the shares $s(z)$:

$$Z = \lambda \int_{\underline{z}}^{\bar{z}} z s(z) dz \quad (40)$$

Equation (40) uses the fact that capital market clearing again implies $1 = \lambda(1 - S(\underline{z}))$.

To measure inequality, I use the Gini coefficient, defined at time t by

$$G_t = 1 - 2 \int_0^1 S_t[F^{-1}(x)] dx \quad (41)$$

In the context of this model, the Gini coefficient measures the discrepancy between $F(z)$, which determines the incidence of each type z in the population, and the wealth shares $S(z)$, which measure the distribution of wealth among these types. Visually, the Gini measures the area between the Lorenz curve, which is constructed by plotting the cumulative wealth shares S against the population shares F . Figure 1 illustrates the Lorenz curve; the Gini coefficient is $1 - 2B$, where B is the shaded area. This measure is convenient, as it can be calculated in the steady state and along transition paths using the wealth shares directly, without needing to infer the underlying wealth distribution.

2.5.1 DETERMINANTS OF WEALTH INEQUALITY

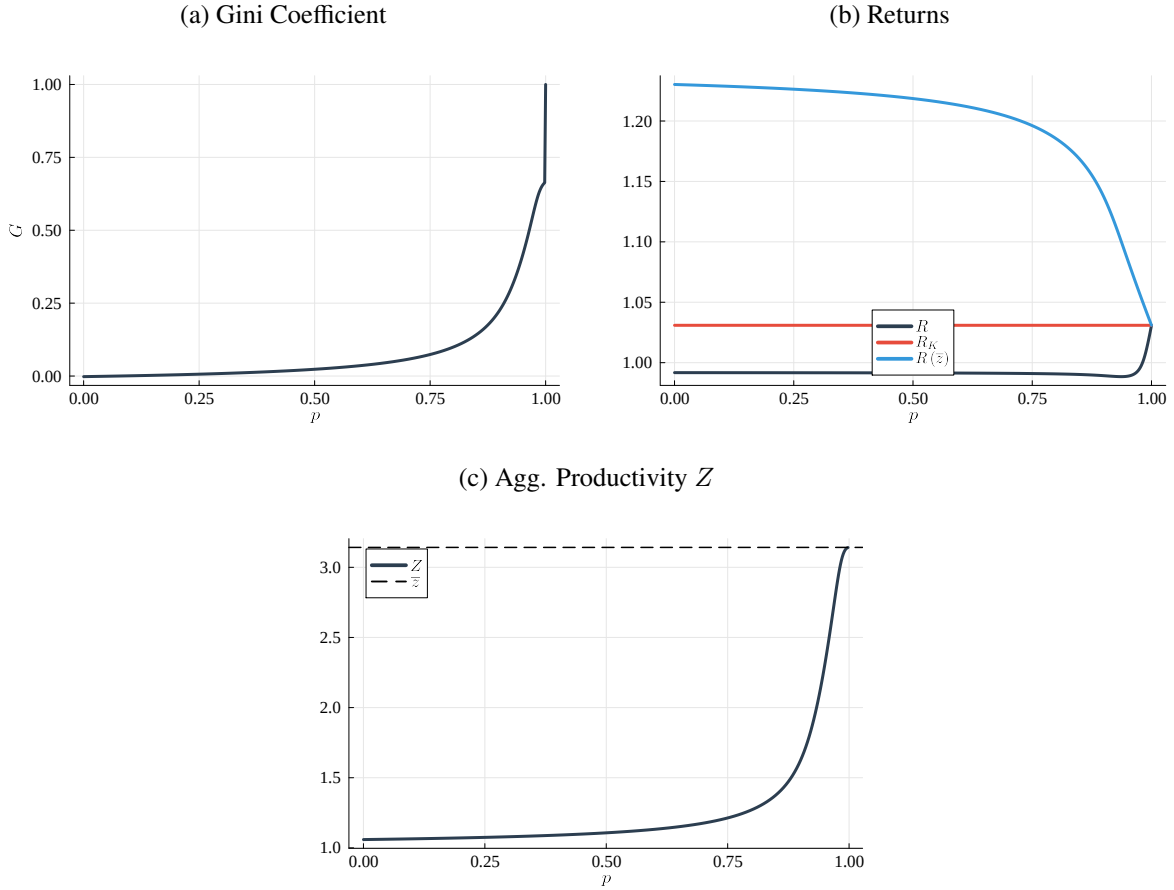
Equations (34)-(36) show the forces that give rise to the steady state wealth distribution. First and foremost among these forces is p , the persistence in returns. At $p = 0$, productivity shocks are IID: each period, entrepreneurs draw a new z_{it+1} that is independent of their current productivity z_{it} . When this is the case, $s(z) = f(z)$: the wealth shares are equal to the marginal density, the Lorenz curve aligns with the 45° line, and the Gini coefficient is zero. In this case, returns and wealth are uncorrelated: entrepreneurs who receive a high productivity shock today are unlikely to receive another high shock tomorrow, and are thus unable to

accumulate wealth from a series of shocks.

At the other extreme, if $p = 1$ then return heterogeneity is permanent. When this is the case, the wealth distribution will be $s(z) = \delta_z(\bar{z})$, where $\delta_z(\cdot)$ is the Dirac measure. In the long run, entrepreneurs of type z end up holding all of the wealth in the economy. As a result, the long-run risk-free rate and return to capital will equate with the returns of entrepreneurs of type $\bar{z}$, $R_K = R = R(\bar{z})$

For intermediate values of persistence, inequality is strictly increasing in p . Figure 2 shows how steady state inequality, returns, and productivity vary in the autocorrelation p . Panel 2a demonstrates that the relationship between inequality and persistence is convex in p . Panels 2b and 2c, meanwhile, shows how aggregate productivity and returns change in persistence. As $p \rightarrow 1$, a greater and greater share of aggregate wealth is held by the top type $\bar{z}$, and the risk-free rate R and the return $R(\bar{z})$ earned by these types converge to $R_K = \beta^{-1}$, while aggregate productivity converges to $\bar{z}$.

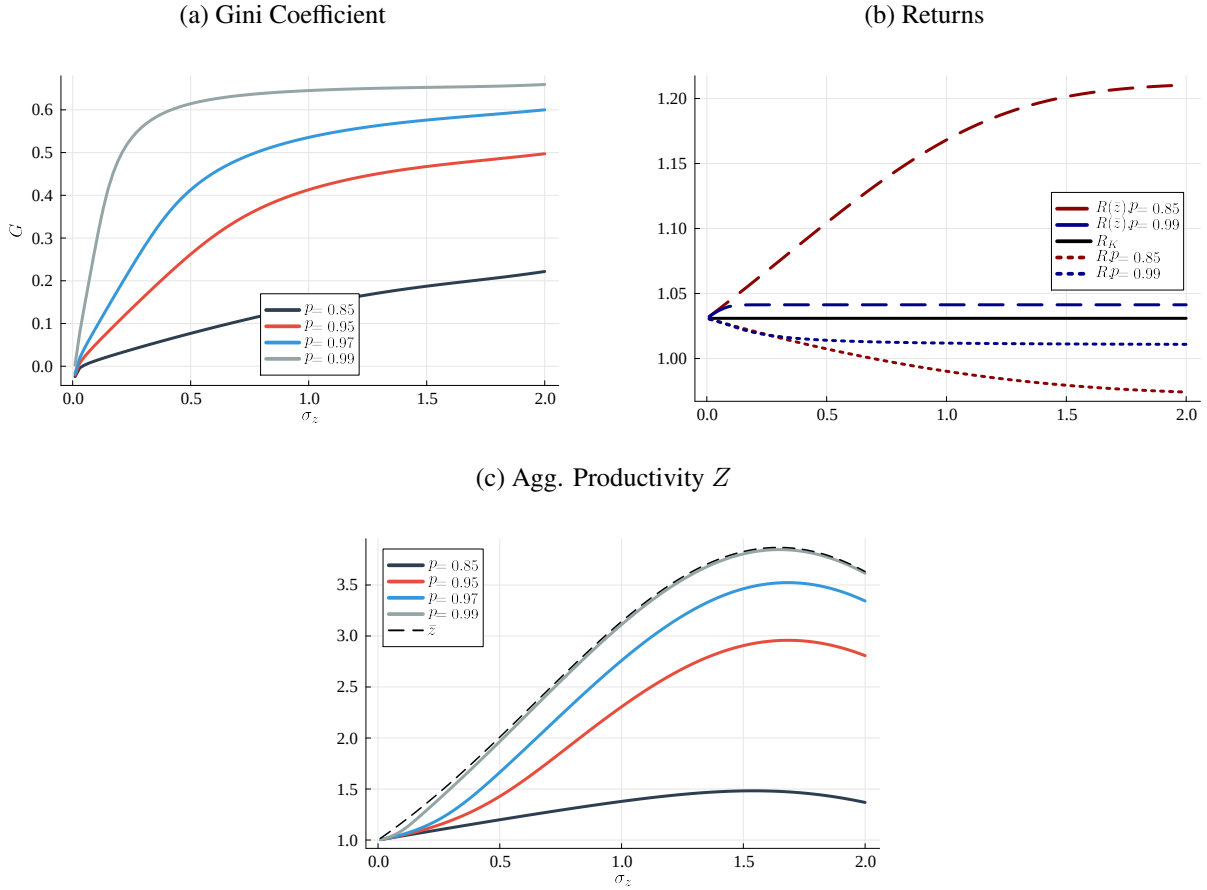
Figure 2: Steady State Comparative Statics in p



Wealth inequality in the steady state is also determined by the *size* of idiosyncratic return shocks, as measured by the standard deviation σ_z of the distribution $F(z)$. Figure 3 shows how wealth inequality, returns, and productivity vary with σ_z across levels of persistence p . At the extreme of no variance, the model once again collapses to one with a representative producer with productivity $z = 1$. Increases in the variance of the idiosyncratic shocks z from this extreme increase the level of wealth inequality, as heterogeneity among

households becomes more salient and their wealth separates over time. However, as is clear from Figure 3, the marginal effect of increases in σ_z on inequality diminishes rapidly, and the overall effect of increases to the variance in shocks depends heavily on how persistent the shocks are. Intuitively, beyond a certain point increasing the size of the largest shocks can only increase wealth inequality by a small amount if these shocks do not persist over time.

Figure 3: Steady State Comparative Statics in σ_z

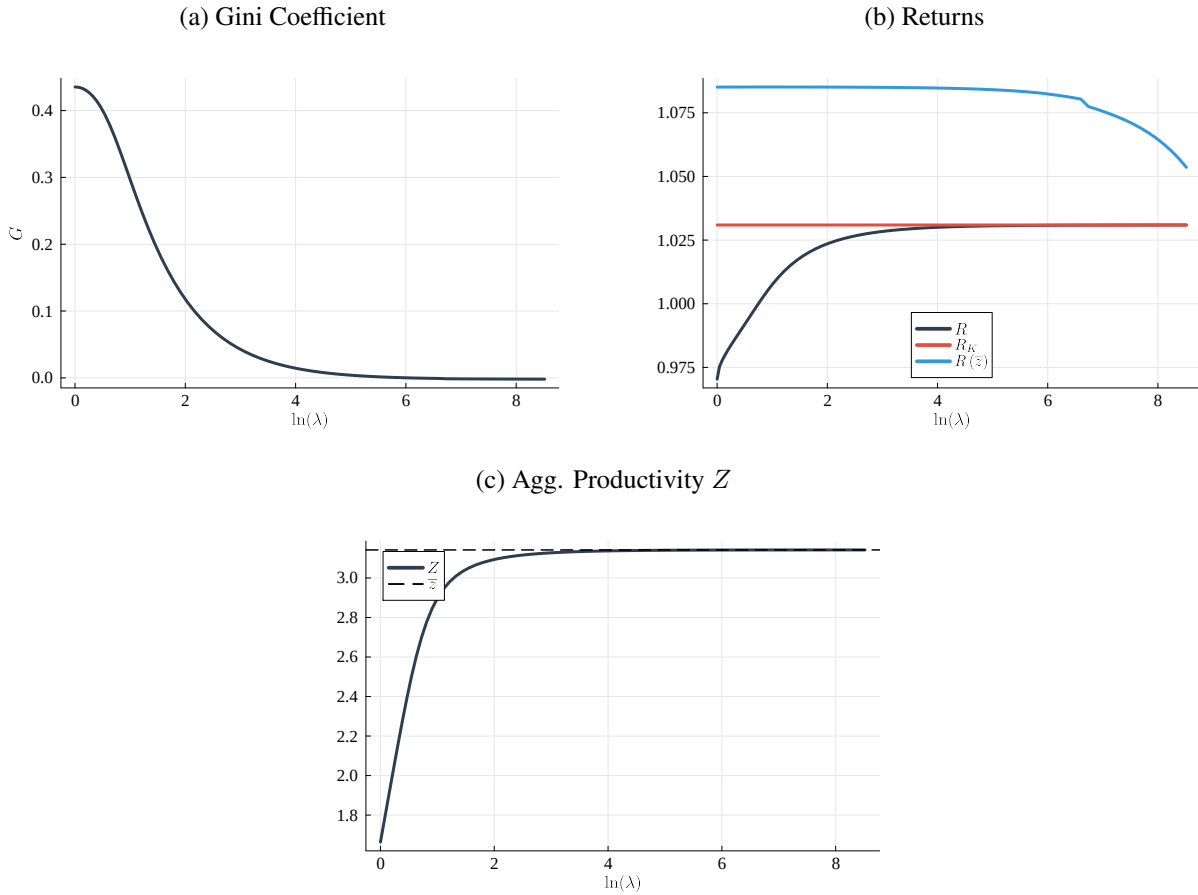


An increase in the collateral constraint λ , meanwhile, *decreases* wealth inequality while *increasing* productivity. Whereas an increase in persistence undoes financial frictions via wealth accumulation, an increase λ reduces misallocation by increasing the scope for *within-period* wealth transfers via the capital market. At the limit of $\lambda = \infty$, entrepreneurs are unconstrained in their ability to take on leverage, and can borrow as much as the market will bear. In this case, the real interest rate is again bid up to the marginal product of the highest-type entrepreneurs, $\bar{z}$, and the economy once again attains its first-best productivity.

Figure 4 shows how productivity, returns, and wealth inequality are affected by an increase in the quality of financial markets, modeled as an increase in λ . Here again, Z is increasing in λ , and returns display a similar pattern, converging to the return of the high types $R(\bar{z})$. The wealth distributions underlying these changes, however, are markedly different. An increase in λ benefits low and high-productivity agents, shifting wealth away from the middle. Intuitively, a loosening of credit constraints allows for high-type entrepreneurs

to borrow more capital from inactive entrepreneurs ($z < \bar{z}$). This has the effect of increasing the risk-free real rate: in the absence of capital market frictions ($\lambda = \infty$), the return to outside savings is equated with the marginal product of the highest types $\bar{z}$, who undertake all of the investment. The increase in the risk-free rate benefits types who do not invest, who now earn a higher return on their savings. The increase in λ also benefits high- z types, who still earn excess returns, but are now able to take on greater leverage to increase their capital income. Looser financial frictions shift wealth away from those with z in the middle of the support, near the cutoff. These types earn small excess returns, and to them, the benefit of looser capital markets is undone by the concomitant increase in the cost of external financing.

Figure 4: Steady State Comparative Statics in λ



The discussion here presages the importance of the wealth distribution in determining the effects of monetary policy. As will become clear in Section 3, different causes for an increase in wealth inequality imply different responses to a policy change: it matters whether high productivity entrepreneurs accumulate capital over time, or borrow it on spot markets. The wealth distribution offers a way to disentangle whether aggregate returns and productivity are being driven by credit markets or by wealth accumulation, and thus what sort of response to policy we may expect.

3 EFFECT OF A MONETARY SHOCK

Here, I consider the effect of an unanticipated change in the stance of monetary policy. I assume that, prior to the shock, the economy is in its long-run, zero-inflation steady state as outlined in Section (2.5). Then, at time $t = 0$, there is an unanticipated innovation $\nu_0 < 0$ to the Taylor rule:

$$i_{t+1} = \bar{r} + \phi_\pi \pi_t + \nu_t$$

The shock ν_0 decays at rate ρ_ν , so $\nu_t = \rho_\nu \nu_{t-1}$. Although agents do not anticipate the initial shock ν_0 , they understand that it will decay according to the process above, and thus for $t > 0$ we return to a *perfect foresight* equilibrium, where there are no further aggregate shocks and agents perfectly anticipate the evolution of all aggregate variables. For a given variable X_t , I denote by $\hat{X}_t \equiv \ln X_t - \ln X$ its log-deviation from steady state.

I begin by focusing on changes in the distribution of wealth, as measured by changes in the wealth shares $\hat{s}_t(z)$ across the space of types z . In period 0, when the shock hits, prices rise, so $\pi_0 > 0$. The rise in prices increases aggregate returns to effective capital, and the increase is linear: $\hat{\omega}_0 = \xi \pi_0 > 0$, where ξ depends only on exogenous parameters. Intuitively, increased demand for consumption and capital goods pushes up prices. Due to nominal rigidities, not all retailer firms can adjust their prices, and must instead adjust the quantity of goods x_t that they purchase from entrepreneurs. This increase in demand for entrepreneurial goods pushes up marginal costs ($\hat{p}_{tx} > 0$), which increases entrepreneurs' return to effective capital.

The following Lemma describes how these changes manifest in changes to wealth shares $\hat{s}_1(z)$ in the period immediately following the monetary policy shock:

Proposition 3. *The changes in wealth shares at $t = 1$ immediately following the shock are*

$$\hat{s}_1(z) = \begin{cases} p\beta R(z) \left\{ \frac{\lambda-1}{R(z)} \pi_0 + \hat{\omega}_0 \left(\frac{\lambda\omega z}{R(z)} - \frac{\omega Z}{R_K} \right) \right\} & z > \underline{z} \\ p\beta R \left\{ -\frac{1}{R} \pi_0 - \hat{\omega}_0 \frac{\omega Z}{R_K} \right\} & z \leq \underline{z} \end{cases} \quad (42)$$

where $\hat{\omega}_0 = \xi \pi_0$.

The proof can be found in Appendix A.3. Equation (42) reveals two channels by which active entrepreneurs benefit from these higher prices. The first channel relates to the change in inflation π_0 , and is known in the literature as the ‘‘Fisher’’ channel (see, e.g., Auclert, 2019). This channel is familiar: the rise in inflation devalues nominal debts, redistributing real assets from lenders to borrowers. The difference in my model is that this channel now benefits high-productivity entrepreneurs ($z > \underline{z}$) – who are borrowers – at the expense of lower-productivity lenders.

The second channel relates to rising profit margins, $\hat{\omega}$. The increase in demand leads to a rise in aggregate profit margins $\hat{\omega}_0$, which benefits active entrepreneurs, who produce in the period of the shock ($t = 0$). Intuitively, the Fisher channel redistributes *existing* wealth (carried forward from the period before the shock), while the profit channel dictates that *newly created* wealth in the expansion is distributed in a pattern that favors higher-productivity entrepreneurs. The following Lemma – the proof of which may also be found in Appendix A.3 – describes which types z benefit from this channel:

Lemma 4. Let $q \equiv \omega Z / R_K = 1 - \beta(1 - \delta)$, which is constant. If

$$r + \delta > \frac{(1 - \delta) q \xi - (\lambda - 1)}{(\lambda - q) \xi} \quad (43)$$

Then all active types $z > \underline{z}$ increase their wealth shares following the shock.

Lemma 4 defines a threshold for the user cost of capital $r + \delta$ that determines whether *all* active types $z > \underline{z}$ benefit from the shock ($\hat{s}_1(z) > 0$), or just those above a threshold $z^\dagger$. Intuitively, this cutoff operates as follows. If capital is sufficiently cheap, then $\underline{z} < z^\dagger$: there will be a mass of types $z \in [\underline{z}, z^\dagger]$ who are active in the steady state, but whose wealth shares *fall* following the shock, as the growth in the aggregate return to capital outstrips their their own returns following the shock. If, on the other hand, outside capital is sufficiently expensive, these marginal types will be crowded out of investing, and $z^\dagger$ will fall below the active cutoff $\underline{z}$. In this scenario, following the shock all active types will see their returns grow faster than the aggregate return on capital, and thus all active types will benefit: $\hat{s}_1(z)$ for $z > \underline{z}$.

To shed further light on how this profit channel operates, and how its benefits are tilted towards higher-productivity entrepreneurs, note that the immediate change in wealth share for active entrepreneurs ($z > \underline{z}$) can be written as

$$\hat{s}_1(z) = p\beta R(z) \left\{ \varepsilon_{R(z),r} \pi_0 + \hat{\omega}_0 (\varepsilon_{R(z),\omega} - \varepsilon_{R_K,\omega}) \right\} \quad (44)$$

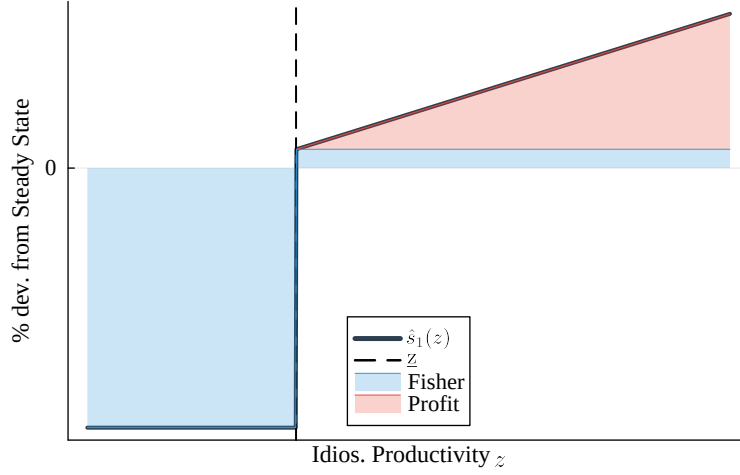
The extent to which active entrepreneurs benefit from the two redistributive channels depends on the elasticity of their returns to aggregate prices. Active entrepreneurs' wealth is increased by inflation – which devalues their debts – in proportion to the elasticity of their returns to the risk-free rate r . Similarly, active entrepreneurs benefit from increases in profits $\hat{\omega}_0$ in proportion to the elasticity of their returns to aggregate profits in the steady state. Here, the relevant measure is the *excess* elasticity $\varepsilon_{R(z),\omega} - \varepsilon_{R_K,\omega}$, which measures how elastic their returns are to ω in excess of the elasticity of the aggregate return on capital to ω . Idiosyncratic returns – augmented by leverage – imply that active entrepreneurs' returns may be *more* elastic to aggregate profit margins than the aggregate return to capital. As a result, when a monetary expansion pushes profit margins up, these households' returns increase by more than the aggregate return. Their wealth in turn grows faster than aggregate wealth, and so their share of aggregate wealth increases further. This channel is stronger, the higher is a household's type z .

Figure 5 shows the pattern in $\hat{s}_1(z)$ across z in the case where $z^\dagger < \underline{z}$: inactive types experience a uniform percentage decline in their wealth shares as a result of inflation, which devalues the nominal contracts written in the period before the shock. Active entrepreneurs benefit both from the inflationary shock, which devalues their *debt*, and from the increase in aggregate demand, visible in $\hat{\omega}_0$. As seen in Figure (5), this change benefits higher- z entrepreneurs more than lower.

This redistribution raises aggregate productivity, further amplifying the effect on output. The redistribution of wealth toward high-types, who benefit from the shock, raises overall productivity. Linearizing the expression for TFP Z_t in Equation (25) and combining with capital market clearing (26), I have the following lemma:

Proposition 4. The change in TFP following the shock ν_0 can be written as $\hat{Z}_1 = \mathbb{E}_s [(z - \underline{z}) \cdot \hat{s}_1(z) | z > \underline{z}]$.

Figure 5: Wealth Share Change $\hat{s}_1(z)$, Following Shock



The change in productivity can be further decomposed into the effects of the Fisher and Profit channels:

$$\begin{aligned}\hat{Z}_1 &= p\beta\pi_0(\lambda - 1)(Z - \underline{z}) && \text{Fisher} \\ &+ p\beta\pi_0\xi\mathbb{E}_s[(z - \underline{z}) \times (\varepsilon_{R(z),\omega} - \varepsilon_{R_K,\omega})] && \text{Profit}\end{aligned}$$

If $p \in (0, 1)$ and $\sigma_z > 0$, then $\hat{Z}_1 > 0$.

The proof of Proposition 4 can be found in Appendix A.4. This proposition demonstrates that the two channels in Proposition 3 affect productivity by redistributing wealth to active entrepreneurs. The Fisher channel, which shifts assets from lenders to borrowers (entrepreneurs), increases productivity in proportion to leverage: the more levered production is, the larger of an effect this channel has. The increase due to the Profit channel depends on the distribution of excess elasticities $\varepsilon_{R(z),\omega} - \varepsilon_{R_K,\omega}$ among active entrepreneurs, weighted by the weights $z - \underline{z}$ and steady-state wealth shares $s(z)$.

As long as $p \in (0, 1)$, and $\sigma_z > 0$ – that is, as long as equilibrium features persistent differences in returns – then a positive price shock π_0 increases productivity unambiguously through both channels. The contribution of the inflation channel is straightforward: by devaluing the debt owed by active investors, this channel pushes assets towards wealthier households and increases overall productivity. The contribution of the profit channel is also unambiguously positive: though the effect of this channel on the wealth share of a given type z is ambiguous (see Proposition 3), it can be shown that the aggregate effect across active households is such that productivity increases. Intuitively, this fact results from the fact that the profit channel in Equation (42) is increasing in z : the more productive a household is, the greater the elasticity of its returns with respect to the aggregate profit margin ω . As such, the benefits of an increase in profit margins are generically concentrated among investors with returns above the return on capital, implying that this channel increases concentration of wealth among these very high-return households, increasing productivity overall.

By capital market clearing, the cutoff $\hat{z}_t$ must rise as well, in order to keep the mass of wealth above the

cutoff constant. Mechanically, the entrepreneurs who benefit ($\hat{s}_1(z) > 0$) from the change in policy wish to invest a portion of their new wealth, in doing so they bid up the cost of capital and thus crowd out marginal entrepreneurs, who instead switch to lending. It is here that we see the importance of the wealth distribution in determining the overall effect of the change in policy: not only does investment increase (an effect which obtains even with identical entrepreneurs, or IID shocks), productivity also changes as the composition of investment is altered, with the additional investment being carried out by entrepreneurs who are more productive than average.

Turning to output, the linearized production function has the usual form:

$$\hat{Y}_t = \alpha (\hat{Z}_t + \hat{K}_t) + (1 - \alpha) \hat{N}_t \quad (45)$$

This expression for output makes clear the amplification in this model, and the surrounding discussion. If shocks are IID, and wealth and productivity are uncorrelated in the steady state, then $\hat{z}_t = \hat{s}_t(z) = 0$ for all t , and thus $\hat{Z}_t = 0$ as well. In this case, Equation (45) becomes $\hat{Y}_t = \alpha \hat{K}_t + (1 - \alpha) \hat{N}_t$, the standard form with fixed productivity. In this world, there is still amplification through investment: $\hat{K}_t$ will increase, as in e.g. Bernanke et al. (1999), which creates an elevated and long-lived response of output to the policy shift. In the case that $p > 0$, however, the results of Section (3) show that the pattern of redistribution $\hat{s}_t(z)$ is such that $\hat{Z}_t > 0$ for a period following the shock. This creates further amplification of the shock, through an increase in productivity, which also occurs in the data (Christiano et al. 2005, Baqaee et al. 2021). These results further imply that this additional amplification also leads to longer-lasting effects: as the shock fades, entrepreneurs slowly spend down their wealth to return to steady state, leaving both investment and productivity elevated for some time after the shock has faded.

The effect of the steady-state wealth distribution on the economy's response to monetary policy can also be seen through the lens of inflation. The linearized Phillips curve is

$$\pi_t = \kappa_p \hat{p}_{tx} + \beta \mathbb{E}_t \pi_{t+1} \quad (46)$$

where $\kappa_p > 0$ determines the slope of the Phillips curve as a function of the elasticity of substitution and adjustment costs. Clearing in the markets for labor and entrepreneurial goods together imply that

$$\hat{p}_{tx} = \frac{\alpha + \eta}{1 - \alpha} \hat{Y}_t - \frac{\alpha(\eta + 1)}{1 - \alpha} (\hat{Z}_t + \hat{K}_t) \quad (47)$$

Combining the two gives

$$\pi_t = \kappa_p \left\{ \frac{\alpha + \eta}{1 - \alpha} \hat{Y}_t - \frac{\alpha(\eta + 1)}{1 - \alpha} (\hat{Z}_t + \hat{K}_t) \right\} + \beta \mathbb{E}_t \pi_{t+1} \quad (48)$$

Above and beyond the impact of additional investment, the change in productivity $\hat{Z}_t$ —itself a function of redistribution, per Proposition 4—has the effect of *lowering* the Phillips curve, implying less inflation for a given pattern of economic activity. The size of this disinflationary force is driven by the response of $\hat{Z}_t$, and so it inherits from Proposition 4 its dependence on the initial distribution. The larger are the gains in

productivity from redistribution, the larger is the *downward* shift in the Phillips curve.

4 WEALTH INEQUALITY AND MONETARY POLICY

Section 3 shows that in the model economy, a primary means by which monetary policy acts on the economy is through increasing aggregate productivity by redistributing wealth towards entrepreneurs with higher idiosyncratic productivities. The increase in aggregate productivity amplifies the effect of the shock, flattening the Phillips curve and increasing the amount of output generated for a given change in capital and labor. The implication of Section 3, then, is this: the overall effect of a change in monetary policy depends on the degree to which this policy change redistributes assets among households. I show now that the extent of redistribution, and thus the overall effect of the monetary shock, depends on the wealth distribution at the time of the policy change.

To show this dependence, in this section I examine analytically how changes in the persistence p of households' returns – the key driver of wealth concentration in the model economy – determines the economy's response to expansionary monetary policy. I first study how persistence affects the response of the wealth shares $s_t(z)$, and then how the response of overall productivity Z_t varies in p . I show computationally that the same intuition accompanies an increase in the volatility of returns, measured by σ_z . For tractability, I focus here again on partial equilibrium, fixing a response of prices π_0 and then studying the initial effect of this change in prices on the wealth distribution and aggregate productivity.

4.1 EXTREME CASES: ESTABLISHING THE BOUNDS

In studying the effect of wealth inequality on monetary policy, it is clearest to start with the extreme cases of zero inequality and perfect inequality, which correspond respectively to $p = 0$ and $p = 1$. In the IID case, with $p = 0$, the law of motion for the wealth shares $s_t(z)$ given in equation (33) implies that $s_t(z) = f(z)$ for all t and z ; the wealth shares are fixed in time. Although entrepreneurs in this case do earn heterogeneous returns, they are just as likely in the next period to be low productivity as they are to be high, and thus wealth shares are unaffected by returns. As a direct result, aggregate productivity and the cutoff $\underline{z}$ are fixed in time as well, equal to

$$\underline{z} = F^{-1}\left(1 - \frac{1}{\lambda}\right), \quad Z = \lambda \int_{\underline{z}}^{\bar{z}} z \, dF(z) \quad (49)$$

At the other extreme of complete inequality ($G = 1$), which corresponds to perfect persistence $p = 1$, a similar result obtains. Because all of the wealth is held by the highest type, no redistribution is possible, and $\underline{z} = Z = \bar{z}$: only the highest types produce, so aggregate productivity coincides with $\bar{z}$. The upshot is that in both cases, the economy acts as one with a representative producer whose productivity is fixed in time. In

both cases, the equations governing the evolution of the economy are:

$$\hat{\omega}_t = \hat{p}_{tx} + (1 - \alpha) (\hat{N}_t - \hat{K}_t) \quad (50)$$

$$\pi_t = \kappa_p \hat{p}_{tx} + \beta_f \mathbb{E}_t \pi_{t+1} \quad (51)$$

$$\hat{i}_{t+1} = \phi_\pi \pi_t + \nu_t \quad (52)$$

$$\hat{r}_t = \hat{i}_t - \pi_t \quad (53)$$

$$\hat{\omega}_{t+1} = \frac{1}{r + \delta} \hat{r}_{t+1} \quad (54)$$

$$\hat{N}_t = \frac{1}{\alpha + \eta} \hat{p}_{tx} + \frac{\alpha}{\alpha + \eta} \hat{K}_t \quad (55)$$

$$\hat{Y}_t = \alpha \hat{K}_t + (1 - \alpha) \hat{N}_t \quad (56)$$

$$\hat{K}_{t+1} = [1 - \beta(1 - \delta)] (\hat{p}_{tx} + \hat{Y}_t) + \beta(1 - \delta) \hat{K}_t \quad (57)$$

Equation (50) pins down the return to effective capital $\hat{\omega}_t$ as a function of aggregates; equation (54) requires that this return comove with the cost of capital, consistent with a constant cutoff $\underline{z}$. Equations (51) and (52) are the linearized Phillips Curve and Taylor rule, respectively. Equation (55) results from clearing in the labor market. Equations (56) and (57) are the linearized equivalents of the production function and capital accumulation equation.

This system can be solved via the method of undetermined coefficients (see, e.g. Chapter 3 of Galí 2015). However, the key property of its behavior following an accommodative shock ($\nu_0 < 0$) can be inferred without any such solution. Note that when linearized about the steady state, the only aggregate steady-state object which appears is r , the steady state real interest rate. As such, all economies featuring constant wealth shares and a common interest rate r will behave *identically* following a monetary shock, regardless of any differences in their steady states. In particular: assuming that both economies are calibrated to match the US economy, such that they replicate the real interest rate observed in the data⁷, the two economies will respond in the same way to a monetary shock, despite displaying polar opposite levels of wealth inequality. Even without re-calibration, the transition paths in these economies will be very similar, as evidenced by the system in (50)-(57). This equivalence between the two extreme economies lays the foundation for the hump-shaped relationship between inequality and the effect of monetary policy: because the response is the same at the polar extremes of inequality, to the extent that interest rate changes redistribute in the intermediate cases, the effect will be larger there. I turn to these cases next.

4.2 INTERMEDIATE CASES: COUNTERVAILING FORCES

For intermediate values of wealth inequality, the tractability of my model allows for further insights as to the role of wealth inequality in determining the effects of monetary policy. Recall that the extent of redistribution, measured as the change in wealth shares $\hat{s}_1(z)$ immediately following the shock, can be written as

$$\hat{s}_1(z) = p\beta R(z) \left\{ \hat{R}_0(z) - \hat{R}_{0K} \right\} \quad (58)$$

⁷In the case of IID shocks, this can be done by choice of $F(z)$ or λ .

The question that I ask now is how this initial redistribution depends on the persistence p . To do so, I fix a value of π_0 to study how redistribution alters the response of the economy to monetary policy for a given price response. Later, I solve for the full response of the economy, including that of prices, computationally.

The forces of redistribution and reallocation are laid out in the following Proposition:

Proposition 5. *Given a price change π_0 , for $z > \underline{z}$:*

$$\begin{aligned} \frac{d}{dp} \hat{s}_1(z) = & \underbrace{\beta R(z) (\hat{R}_0(z) - \hat{R}_{0K})}_{\text{Term 1}} \\ & + p\beta \times \underbrace{\frac{d}{dp} \left\{ R(z) \cdot (\hat{R}_0(z) - \hat{R}_{0K}) \right\}}_{\text{Term 2}} \end{aligned}$$

Term 1 is positive for $z > \underline{z}$, by construction. Term 2 is negative for $z > \underline{z}$ as long as

$$\frac{|\varepsilon_{\omega,p}|}{\varepsilon_{z,p}} > \Xi \quad \text{where} \quad \Xi \equiv \frac{q(\lambda - 1)\underline{z}}{\lambda(1 - q)z + q(\lambda - 1)\underline{z}} > 0 \quad (59)$$

where $\varepsilon_{\omega,p}$ and $\varepsilon_{z,p}$ are the elasticities of the aggregate return ω and cutoff $\underline{z}$ in p , respectively. Furthermore, $\hat{s}_1(z) \Big|_{p=1} = 0$ but

$$\lim_{p \rightarrow 1} \hat{s}_1(z) = \begin{cases} 0 & z < \bar{z} \\ \pi_0(\lambda - 1)(\beta + \xi q) & z = \bar{z} \end{cases} \quad (60)$$

The proof of Proposition 5 can be found in Appendix A.5. This proposition highlights the countervailing forces that affect redistribution as wealth inequality, driven by changes in persistence, increases. The first term, redistribution, is positive. To understand the intuition behind the redistribution channel, fix a type z and a level of excess returns $\widehat{ER}_0(z) = \hat{R}_0(z) - \hat{R}_{0K}$. Term 1 measures the extent to which this change in excess returns filters into wealth shares: the more persistent are returns, the greater is the scope for agents who earn returns in excess of the return on capital following the shock to *accumulate* from those returns. Term 2, however, is negative for $z > \underline{z}$, and captures the reallocation channel. This term captures, for a given level of persistence (and thus scope for accumulation), what level of excess returns the agents can expect to earn following the shock. To understand this term, recall the comparative statics on steady-state returns in Figure 2b. As wealth concentration increases, the idiosyncratic returns $R(z)$ for active entrepreneurs ($z > \underline{z}$) align with the return on capital. Proposition 5 shows that this property of the steady state also holds along the transition path following a monetary shock: as $p \rightarrow 1$, idiosyncratic returns $\hat{R}_0(z)$ align with the return on capital $\hat{R}_{0K}$, and thus $\widehat{ER}_0(z)$ declines in p .

The inequality in (59) is a restriction on parameters that ensures excess returns shrink in p (Term 2). The excess returns earned by an active type $z > \underline{z}$ following the shock depend on two equilibrium objects: the aggregate return to effective capital ω , and the risk-free rate $r = \omega\underline{z} - \delta$. As established in Appendix A.5, $\omega_p < 0$: the return ω shrinks with persistence. The response of r is not necessarily monotonic, but generally, r is low for low values of persistence and rises with persistence until it coincides with the return on capital at $p = 1$ (see Figure 2b). The key intuition is that excess returns to type z shrink in p so long as

the entrepreneurs' return to capital ωz shrinks to zero faster than the risk-free rate reaches its limit, with factors on each term representing the contribution of each to the excess return. Rearranging this condition, accounting for the fact that ω appears in both terms, results in the condition (59).

Figure 6 illustrates the results of Proposition 5. In each panel, I plot the responses of the variables in question again given a fixed price response π_0 . Panel 6a shows the initial change in the wealth share of the highest type, $\hat{s}_1(\bar{z})$, following the shock across persistence p . As illustrated in Proposition 3, the changes for all of the remaining active entrepreneurs will be bounded above by $\hat{s}_1(\bar{z})$. Panel 6a shows the hump-shaped response of wealth shares, as a result of the redistribution and reallocation channels. Panel 6b shows the derivative $d\hat{s}_1(\bar{z})/dp$ in black, and the two terms in red and blue respectively.

As in Proposition 5, the first term is positive, reflecting the benefit of accumulation that comes from higher persistence. The second term is negative, reflecting the logic of the reallocation channel: further wealth concentration in the steady state reduces the scope for active entrepreneurs to earn returns in excess of the return on capital. Panel 6c illustrates this point directly: analogously to Figure 2b, Panel 6c shows the idiosyncratic return of type- $\bar{z}$ agents immediately following the shock, $\hat{R}_0(\bar{z})$, as well as the return on capital $\hat{R}_{0K}$, for fixed π_0 across p . Note that for a given change in prices, the response of the return on capital $\hat{R}_{0K}$ is fixed and independent of steady-state wealth concentration. As in Figure 2b, we see that as p increases, the return earned by these top entrepreneurs converges to the return on capital, and their difference $\widehat{ER}_0(\bar{z})$ in light blue converges to zero.

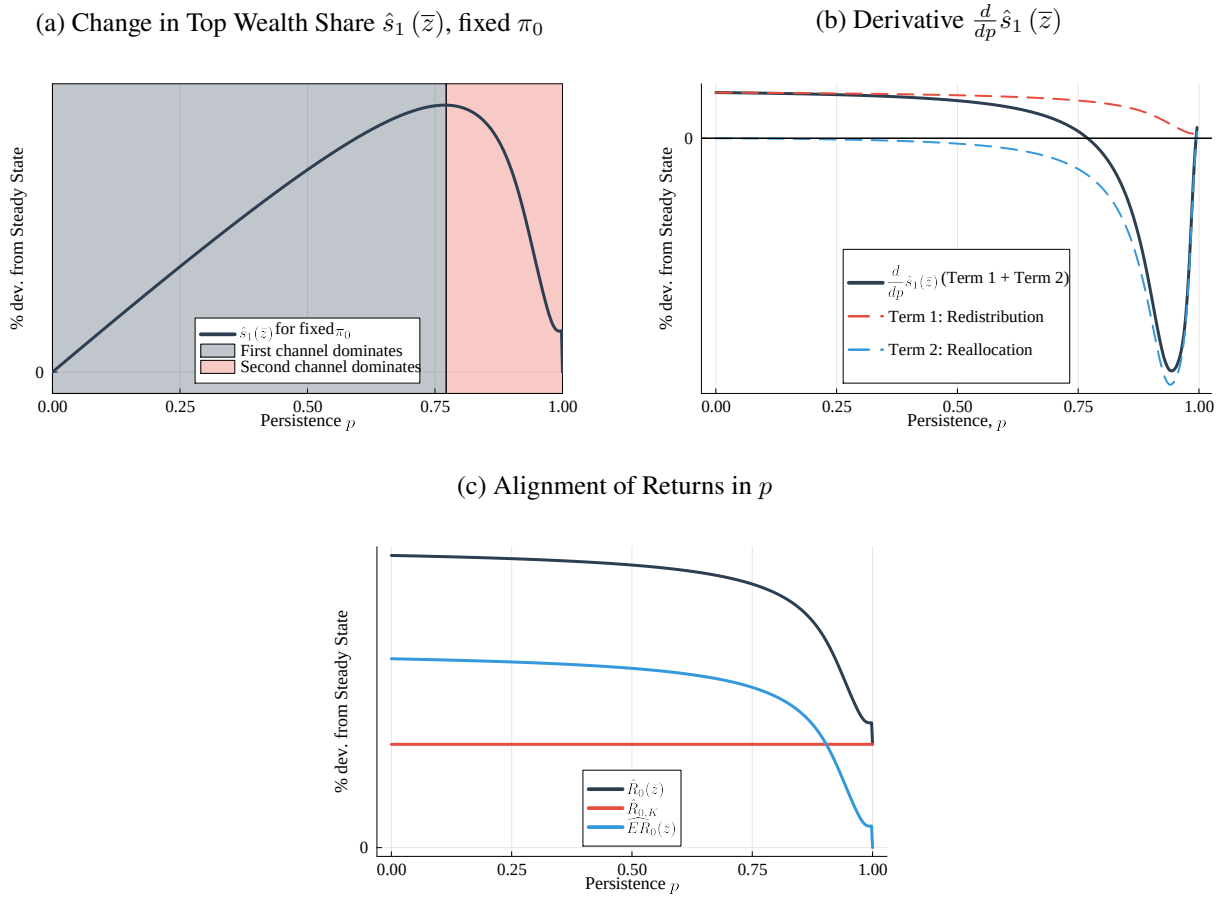
Note that each of the panels in Figure 6 reflect the discontinuity in Proposition 5: for any p arbitrarily close to one, credit markets will be active, and entrepreneurs will borrow up to their debt limit. At $p = 1$, however, the top types hold all of the wealth, and the credit markets vanish — inactive entrepreneurs no longer hold any wealth to lend to the highest types. As such, for any p near but less than one, active entrepreneurs still benefit from the Fisher channel described in Section 3, and thus still enjoy a small amount of redistribution from inflation. At $p = 1$, meanwhile, credit markets are inactive, and the idiosyncratic return of the top types is by definition equal to the return on capital, thus shrinking redistribution to zero.

With the results of Proposition 5 established, the connection between wealth inequality and monetary policy is immediate. Because the magnitude of redistribution changes with wealth inequality, so too does the magnitude of the response of productivity to the shock. The amplification of the shock, and therefore its overall effect on aggregates, then depends on how unequally wealth is distributed to begin with. To pin down this relationship, the following proposition calculates the derivative of the initial change in TFP $\hat{Z}_1$ in p :

Proposition 6. *The derivative of the productivity response $\hat{Z}_1$ in p can be decomposed into the following three terms:*

$$\begin{aligned} \frac{d}{dp} \hat{Z}_1 = & - \left(\frac{dz}{dp} \right) \mathbb{E}_s [\hat{s}_1(z) \mid z > \underline{z}] & 1. \text{ Participation} \\ & + \mathbb{E}_s \left[(z - \underline{z}) \times \frac{d\hat{s}_1(z)}{dp} \mid z > \underline{z} \right] & 2. \text{ Redistribution} \\ & + \mathbb{E}_s \left[(z - \underline{z}) \hat{s}_1(z) \times \left(\frac{1}{s(z)} \frac{ds(z)}{dp} \right) \mid z > \underline{z} \right] & 3. \text{ Reweighting} \end{aligned}$$

Figure 6: Redistribution across Persistence



The first term, Participation, is unambiguously negative. The second term, Redistribution, is positive if

$$|\varepsilon_{\omega,p}| < \frac{V + (Z - \underline{z}) \left(\Delta(Z) + \frac{q(\lambda-1)}{\lambda(1-q)} \underline{z} \varepsilon_{z,p} \right)}{V + (Z - \underline{z}) \left(Z + \frac{q(\lambda-1)}{\lambda(1-q)} \underline{z} \right)} \quad (61)$$

where

$$\Delta(Z) \equiv \frac{\hat{s}_1(Z)}{p\beta\pi_0\xi\lambda(1-q)\omega} > 0$$

and $V \equiv \text{Var}_s[z \mid z > \underline{z}]$. The third term, Reweighting, is positive if

$$|\varepsilon_{\omega,p}| < \frac{1 - p\beta(\lambda - 1)\omega \underline{z} \varepsilon_{z,p}}{1 - p(1 - q)} \quad (62)$$

Proposition 6 – the proof of which can be found in the Appendix – says that the response of aggregate TFP to the monetary shock is *also* subject to countervailing forces in the persistence of entrepreneurs' returns. These forces manifest through three terms, each of which reflects changes to the economic environment as persistence p changes.

The first term, Participation, captures how changes in the cutoff productivity $\underline{z}$ alter the response of productivity to the price shock. Recall from Proposition 4 that the contribution of each type z to the overall change in productivity is weighted by its distance to this cutoff, $z - \underline{z}$. An increase in persistence pushes the cutoff $\underline{z}$ upwards (see Lemma 5 in the Appendix), which decreases the weight that each type receives. Holding fixed both the steady-state wealth shares, and initial redistribution $\hat{s}_1(z)$, this term decreases the response of TFP to a monetary shock. Intuitively, given a starting distribution $s(z)$ and the response of wealth shares $\hat{s}_1(z)$, each type z contributes marginally less to overall productivity with a higher cutoff.

The second term, Redistribution, captures how the response of wealth shares changes with persistence p , and how these changes affect the response of TFP $\hat{Z}_1$. This term can take either sign: it is positive at low values of p , and negative at high values. The inequality in (61) determines the direction of this term. This inequality is the distributional analogue of the single- z term in (59): the term in equation (59) determines whether excess returns are shrinking in p for a single type z , while the condition in (61) determines whether the cumulative force of these downward pulls across the distribution of active investors ($z > \underline{z}$) is sufficient to pull down the response of productivity. It is through this term that the aggregate productivity response $\hat{Z}_1$ inherits the hump shape in the individual shares $\hat{s}_1(z)$.

Finally, the third term captures Reweighting: the effect of the initial steady-state distribution $s(z)$ on the productivity change $\hat{Z}_1$ following the monetary shock. This term drives the results in Matusche and Wacks (2021) and González et al. (2024): under certain conditions, shifting steady-state wealth towards investors with high productivity z implies greater increases in *aggregate* productivity following the shock. The intuition in these models is as follows: recall that to first order, $\hat{Z}_1 \propto \int_{\underline{z}}^{\bar{z}} (z - \underline{z}) \hat{s}_1(z) s(z) dz$. Thus, holding all else fixed, increasing the steady-state weights of high-productivity investors results in a larger productivity gain: these investors control a higher share of aggregate wealth, and thus their increase in wealth share $\hat{s}_1(z)$ feeds through more strongly into aggregate productivity.

As I have demonstrated, however, this intuition is incomplete for two reasons. First, note that this channel

is only positive if the inequality in (62) is satisfied. While this is generally the case, it is not guaranteed to be so. Once again, general equilibrium effects act as counterbalances: if the condition on the elasticities $\varepsilon_{\omega,p}$ and $\varepsilon_{z,p}$ is not satisfied, then an increase in persistence p pushes up the cost of capital by more than it increases high- z types' returns, and their wealth shares *fall*. Second: when inequality increases, all else generally is *not* equal. Increasing the steady-state wealth shares of high- z types necessarily increases the cutoff and, past a certain point, limits the ability of monetary policy to eke out productivity gains by further redistribution. Neglecting these forces – which my model transparently uncovers – may lead one to conclude that increases in wealth inequality unambiguously lead to higher productivity responses, a conclusion which is both untrue, per my Proposition 6, and at odds with the data.

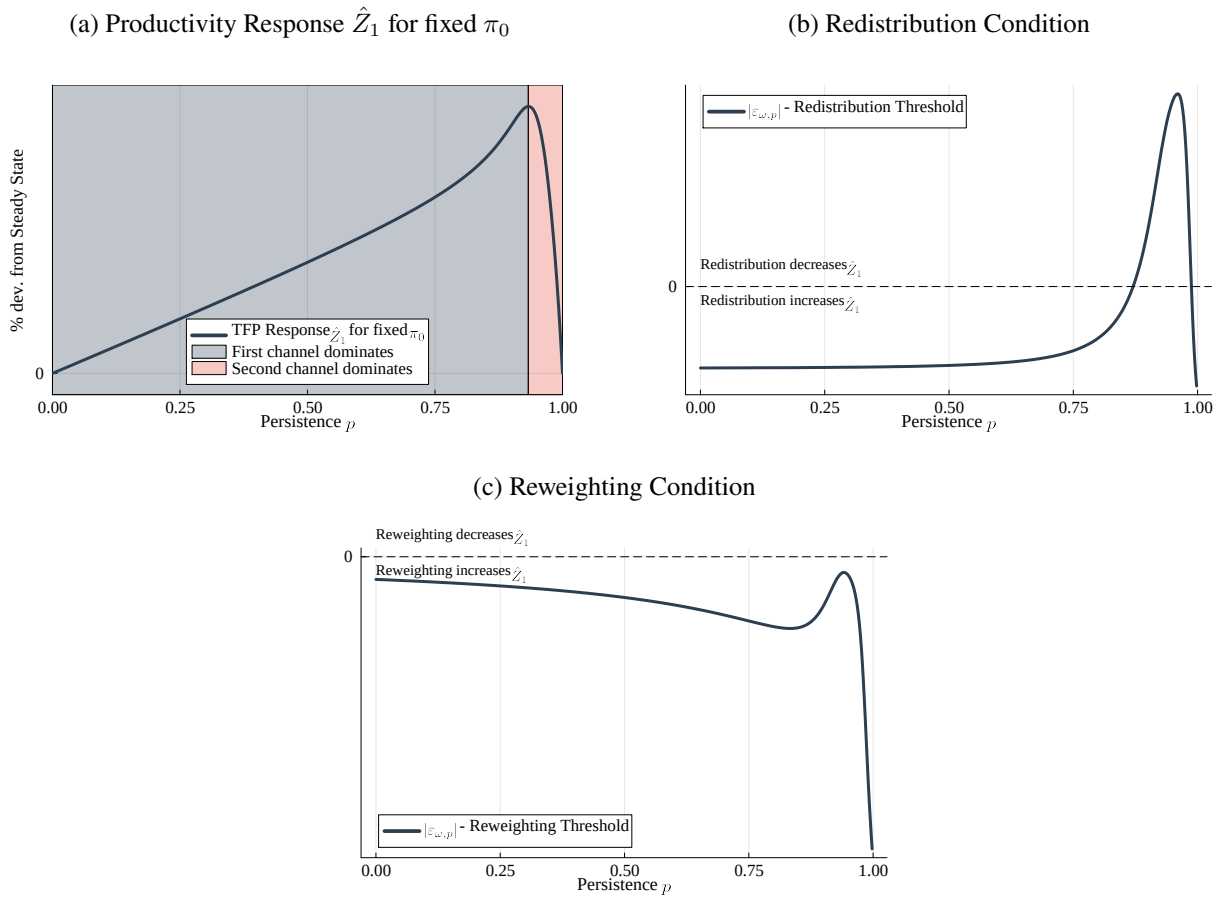
Figure 7 illustrates the response of $\hat{Z}_1$ across persistence p , again for fixed π_0 . Panel 7a shows how the response of TFP varies with persistence, with the blue area indicating the region where $d\hat{Z}_1/dp > 0$ and red indicating where $d\hat{Z}_1/dp < 0$. Panels 7b and 7c show the difference between the elasticity of ω in p , and the thresholds in Equations (61) and (62) respectively.

In Panel 7a, note that the hump-shaped pattern in redistribution in Figure 6a translates to a hump-shaped pattern in the TFP change $\hat{Z}_1$. Aggregate TFP does not respond to a monetary shock at $p = 0$ and $p = 1$, and the magnitude of its response peaks between these two values. Notably, though, the TFP response of peaks at a higher value of p than does the response $\hat{s}_1(\bar{z})$ in Figure 6a. There are two reasons for this discrepancy. First, note that in Panel 7b, for most persistence values the condition in (61) holds, and changes in redistribution as p increases feed through to *larger* changes in TFP. This figure illustrates the difference between conditions (59) and (61): the former determines whether increases in p weigh negatively on the wealth share response of a single type z , whereas the latter determines whether the increase in p weighs on the response of TFP, which weights all of the wealth changes among active investors by $(z - \underline{z})s(z)$. The key insight is this: although the wealth share changes $\hat{s}_1(z)$ shrink to zero as $p \rightarrow 1$ for all active types, they shrink more slowly for higher z types. As such, even as redistribution begins to shrink for types near $\bar{z}$, it remains elevated relative to types with lower productivities, keeping the response of $\hat{Z}_1$ elevated for longer.

Additionally, Panel 7c shows that for this set of parameters, the condition in equation (62) holds for all p and changes in the baseline distribution $s(z)$ as p increases further prop up the TFP response — put differently, the Reweighting term in Proposition 6 is positive. This is the benefit of higher wealth inequality among firms in Baqaee et al. (2021), and households in Melcangi and Sterk (2024): even though redistribution itself is shrinking, the redistribution that does occur is better targeted towards perennially high-return investors, who in turn exert a larger weight on the total response of $\hat{Z}_1$. However, Figure 7 demonstrates that these benefits fade away for high values of wealth inequality: as wealth concentration nears its maximum, the response of TFP to monetary policy begins to fade to zero. Note that unlike the redistribution terms $\hat{s}_1(z)$, $\hat{Z}_1$ fades to zero smoothly: although the Fisher channel buoys wealth share changes even as excess returns disappear near $p = 1$, the weights $z - \underline{z}$ that determine the productivity response shrink as the cutoff $\underline{z}$ approaches the upper bound $\bar{z}$.

As a result of the hump-shaped pattern of the TFP response $\hat{Z}_1$ in Figure 7, the *overall* response of aggregate productivity to an initial change in prices π_0 is hump-shaped in the persistence of households' returns, and thus hump-shaped in wealth inequality. In Section 5, I demonstrate that these partial-equilibrium

Figure 7: Productivity Change across Persistence



forces survive general-equilibrium effects, and the response of the aggregate economy to monetary policy is hump-shaped in the persistence of households' returns. Put simply, the degree of wealth inequality present at the time of a change in the policy rate determines how much redistribution the policy rate creates, and thus how much the response is amplified by the resulting change in aggregate productivity. The results here also imply that beyond a certain point, increases in wealth inequality *dampen* the efficacy of monetary policy.

4.3 EFFECT OF INEQUALITY THROUGH VOLATILITY

Sections 4.1 and 4.2 establish how the aggregate effects of monetary policy vary with the persistence of returns p . While persistence is the primary driver of steady-state wealth inequality in the model, it is not the only one: along with the autocorrelation in returns, wealth inequality increases in σ_z . The standard deviation of $f(z)$, the renewal distribution, this parameter determines the size and volatility of the return shocks that agents experience. Here, I demonstrate that the same forces studied in Section 4.2 manifest as the size of shocks increases: the effects of monetary policy are hump-shaped in σ_z .

Figure 8 shows the response of the top-type wealth share $\hat{s}_1(z)$ and aggregate productivity $\hat{Z}_1$, again for a fixed price change π_0 , for a few values of persistence p . These plots are analogous to Figures 6a and 7, but with σ_z varying for the given persistence p . Recall that in the steady state, the effect of σ_z on wealth inequality is highly dependent on p : and increase in return volatility only manifests in greater inequality to the extent that households are able to keep their high return shocks over time.

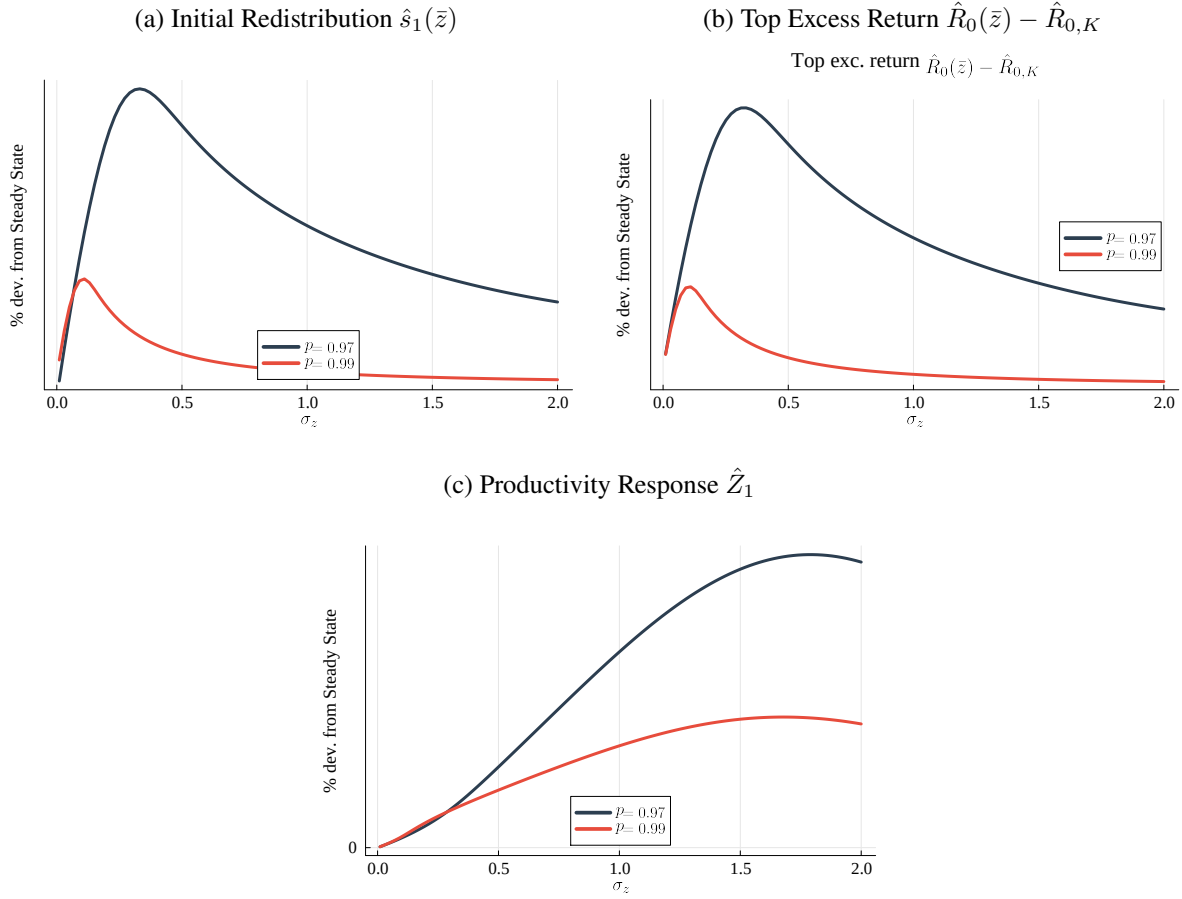
This dependence that plays out in the steady state also influences the economy's response to a monetary shock. In the left-hand panel of Figure 8, increases in σ_z from zero increase the response of the wealth shares. This result is easily interpretable: $\sigma_z = 0$ corresponds to identical returns across the population, in which case aggregate productivity is again unresponsive to monetary policy. As such, increases in σ_z from this benchmark represent another instance of the monetary redistribution being *well-targeted*: the beneficiaries of the initial shock now truly have higher returns, and can beat the return on capital and increase their wealth share. Again, crucially, this effect depends on returns being persistent.

As σ_z continues to grow, however, the hump shape in $\hat{s}_1(z)$ returns: past a certain point, increases in wealth inequality through greater volatility in returns *dampen* the size of redistribution following the monetary shock. This dampening can be seen in Figure 8a: beyond a given level – which depends on p – further increases in σ_z dampen the response of $\hat{s}_1(\bar{z})$. The reason is the same as before, and is in fact quite general. Recall that agents accumulate wealth $a_{i,t}$ according to $a_{i,t+1} = \beta R_t(z_{i,t})a_{i,t}$. As σ_z increases, agents receiving the highest z shocks accumulate wealth faster and faster relative to their lower- z counterparts, resulting in them holding a larger share of wealth in the steady state, per section 2.5. As this concentration increases, the second force discussed in Proposition 5 begins to dominate: because these extraordinarily high-return investors hold so much of the aggregate capital stock, their returns begin to closely track the return on capital, which reduces their wealth share gain following the shock.

As it does when p increases, the hump-shaped pattern in redistribution as σ_z grows feeds through to productivity, resulting in a pattern of initial productivity changes $\hat{Z}_1$ that is hump-shaped in volatility. Again, the same forces are at work: for increases in σ_z past a given level, the returns to active investors begin to compress towards the return on capital. This compression – again resulting directly from increased

concentration – implies that following the shock, the wealth of these investors grows at a similar rate to overall capital, holding their wealth shares relatively more fixed. As a result, beyond a given level of σ_z , the productivity gains from redistribution shrink. Notably, the value of σ_z at which these responses peak changes with persistence p . In the next section, I calibrate my model to the data and find that the model attributes the increase in inequality since the 1970s to an increase in both return persistence p and volatility σ_z , and that both increases have the effect of dampening the overall effects of monetary policy.

Figure 8: Redistribution and Productivity Change σ_z



5 QUANTITATIVE RESULTS

In Sections 3 and 4, I study theoretically the response of my economy to monetary policy, and measure how that response depends on wealth inequality. Here, I calibrate my model to match moments of the joint distribution of wealth and returns in the US economy as of 2022, to assess the quantitative relevance of the channels studied above. To begin, I discuss my strategy for calibrating my model to the data. Then, with the calibrated model in hand, I ask a few questions about how the model speaks to the data.

First, I show that this model captures the response of the economy to monetary policy fairly well; in particular, it captures movements in TFP and inequality resulting from monetary easings. Then, I show that

Table 1: Quarterly Calibration, 2022

<i>Parameter</i>	<i>Value</i>	<i>Source</i>
β	0.99	Avg. return to wealth (see text)
δ	0.021	K/Y of 12
η	2	Labor elasticity (standard)
λ	1.43	Debt/Assets (FRB)
ε	10	Avg. Markup 11%
θ	90	Slope of Phillips Curve 0.1
α	0.4	Capital share 36% (standard)
ϕ_π	1.5	Standard value
ρ_ν	0.6	Standard value
μ_w	0.4	Lorenz Curve (SCF)
p	0.98	Wealth Gini of 0.8 (SCF)
σ_z	2.75	Cross-section of returns (see text)

the degree of amplification resulting from the redistribution in assets is substantial, implying that this is a meaningful channel of monetary policy to capture. My calibrated model also implies a markedly different relationship between output and inflation than in standard RANK and HANK models, again arguing that heterogeneity in household returns is an important source to capture. Finally, I show that my calibration suggests that the increase in wealth inequality since the 1970s can partially account for the decrease in the efficacy of monetary policy that has been documented in the empirical literature over the same time period.

5.1 CALIBRATION

The time period in my model is one quarter. My parameters are summarized in Table 1. To begin, in my model the wealth-weighted average return to wealth in the steady state is equal to $1/\beta$, and so I set this to match the average return in Fagereng et al. (2020), approximately 0.9 percent quarterly. Depreciation is then set so that the quarterly capital stock to output ratio is 12. Note that in my model, this ratio is given by

$$\frac{K}{Y} = \frac{\alpha p_x}{\beta^{-1} - (1 - \delta)}. \quad (63)$$

As a result of my assumption on workers' preferences, workers supply labor with elasticity $1/\eta$. McClelland and Mok (2012) report an estimate of this elasticity of about 0.5, implying $\eta = 2$. These households hold no wealth, and so I assume that they make up 40% of the population ($\mu_w = 0.4$), corresponding to the fact that in the SCF, the bottom 40% of households by wealth cumulatively hold approximately zero net worth.

For λ : entrepreneurs in my model borrow a proportion $\lambda - 1$ of their assets in debt, meaning for active entrepreneurs, $d/k = (\lambda - 1)/\lambda$. I therefore set λ so that firms in equilibrium have a debt to assets ratio of 0.44, which is the long-run average debt-to-assets ratio as measured by the Federal Reserve Board's Financial Accounts of the United States for corporate and non-corporate businesses combined. I choose this particular leverage ratio to capture the debt that backs productive business assets, which are a primary asset class among wealthy households with high returns.

The Phillips curve parameters ε and θ follow Kaplan et al. (2018), and imply a Phillips curve slope of 0.1, as in Schorfheide (2008). This choice is intentional: to measure how productivity gains translate to prices, it must be that in my economy a change in marginal costs has the same effect on prices as in a standard HANK model. This way, any difference in the Phillips curve between my model and those in the literature come through the changes in marginal costs themselves. The capital loading factor α is then set so that the capital share αp_x is 0.36, a standard value. The weight on inflation in the Taylor Rule ϕ_π and the persistence ρ_ν are assigned standard values from the literature.

What remains are the persistence of entrepreneurs' returns p and the distribution of types $F(z)$, parameterized by σ_z . These two parameters jointly target two moments: a wealth Gini of 0.8 – the average in the SCF data from 1970 to 2022 – and the wealth-weighted, pre-tax cross sectional standard deviation of returns to wealth. Per Fagereng et al. (2020), this dispersion is about 1.95% at a quarterly frequency.

5.2 IMPULSE RESPONSES IN THE CALIBRATED MODEL

Here, I discuss the response of my economy to a 100bp monetary shock. To calculate the full impulse response, I use a computational strategy, which I outline in Appendix A.8. To begin, Figure 9 shows the responses of two variables of particular interest in my model: aggregate TFP $\hat{Z}_t$, and the wealth Gini coefficient $\hat{G}_t$. The impulse responses in red are computed from my model; those in black are estimated using Jordà (2005) local projections.

To estimate the empirical response of total factor productivity, I use data on aggregate TFP from the Federal Reserve Bank of San Francisco and follow Baqaee et al. (2021) in estimating the following specification:

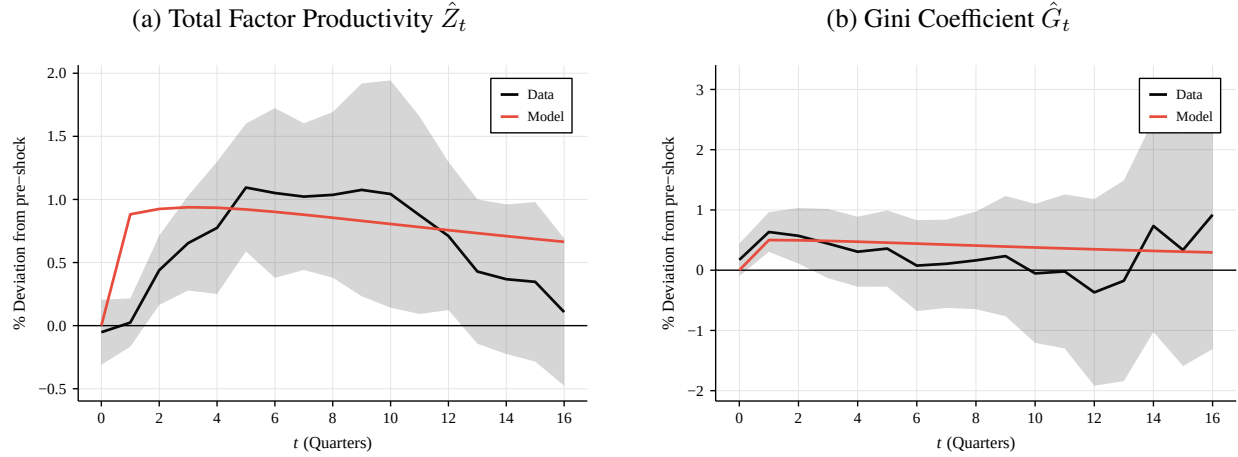
$$z_{t+h} = A + \sum_{k=0}^4 B_{h,k} \nu_{t-k} + \sum_{j=1}^4 C_{h,j} z_{t-j} + \epsilon_t \quad (64)$$

Here, the terms ν_t are monetary shocks – identified via the narrative approach in Romer and Romer (2004) and extended through 2012 by Wieland and Yang (2020) – and z_t is total factor productivity for the U.S. economy. The impulse responses, then, are the $B_{h,0}$ terms, which measure the impact of a shock in the current period on h -period ahead forecasts of TFP.

To estimate the response of inequality, I use data on wealth shares from the Federal Reserve Board's Distributional Financial Accounts (DFA) for the U.S. This dataset contains a quarterly time series on wealth shares for the bottom 50% of the wealth distribution, as well as top shares for selected percentile bins. Using these I compute the Gini coefficient by reconstructing the Lorenz curve using the procedure in Medlin (2023). One can verify that the estimates of the Gini coefficient constructed from the DFA data track those implied by the triennial SCF waves in both magnitude and time-series pattern. With the quarterly Gini estimates in hand, I estimate the response of the Gini coefficient to the identified monetary shocks using the same specification as in (64).

At its peak, TFP $\hat{Z}_t$ in Panel 9a increases by about 1% relative to the steady state — a similar magnitude to its response in the data. On inequality $\hat{G}_t$, the model fit is similarly good: the Gini coefficient increases at its peak by about 0.5%, about four-fifths the size of the increase in the Gini Coefficient that I estimate in the

Figure 9: Impulse Responses to 100bp monetary shock



Note: grey shaded areas indicate 90% confidence intervals using Newey-West standard errors.

data, as well as that in Medlin (2023). In total, Figure 9 suggests that despite being relatively parsimonious, the model can accurately capture the responses of both wealth inequality and productivity to a monetary shock — responses that are not captured in standard HANK models.

Figure 10 shows the responses of output and investment in the calibrated model. Unsurprisingly, both increase, and investment by more than output, as is the case in the data. Figure 11, meanwhile, shows the remaining impulse responses. In particular, Panel 11a shows the response of the overall price level π_t , as well as the response of entrepreneurs' price $\hat{p}_{tx}$ and profit margin $\hat{\omega}_t$. Here, both the price that entrepreneurs earn on their goods $\hat{p}_{tx}$ and their profit margin per unit of effective capital $\hat{\omega}_t$ both rise in response to the increase in demand. As a result, then, the overall price level rises as well. Panel 11b, meanwhile, shows the impulse responses of the nominal and real interest rates $\hat{i}_t$ and $\hat{r}_t$ respectively, as well as the response of the aggregate return on capital $\hat{R}_{tK}$. The real interest rate falls as the monetary authority loosens, as intended, and then rises following the subsequent tightening that the policymaker enacts to counter inflation. As a result of the rise in prices and returns, as well as the redistribution that follows, the return on capital $\hat{R}_{tK}$ rises as well.

5.3 AMPLIFICATION

Here, I provide two arguments as to the empirical significance of the redistributive channel of monetary policy. To begin, Figure 12 assesses the effect of redistribution on output $\hat{Y}_t$ and investment $\hat{I}_t$. For each variable, I plot two impulse responses to the same 100bp monetary shock as above. In blue, I plot the full impulse response, allowing for redistribution, which is identical to that in Figure 10. In red, meanwhile, I plot the impulse response of an identical economy with redistribution shut down: I begin from the same steady state, but fix wealth shares $s_t(z)$ following the shock. Clearly, the redistributive channel exerts a strong influence on output in particular: the increase in productivity that results from shifting assets towards higher-productivity entrepreneurs implies that over the life of the shock, the cumulative change in output is about fifty percent higher than the world without redistribution.

Figure 10: Impulse Responses to 100bp monetary shock

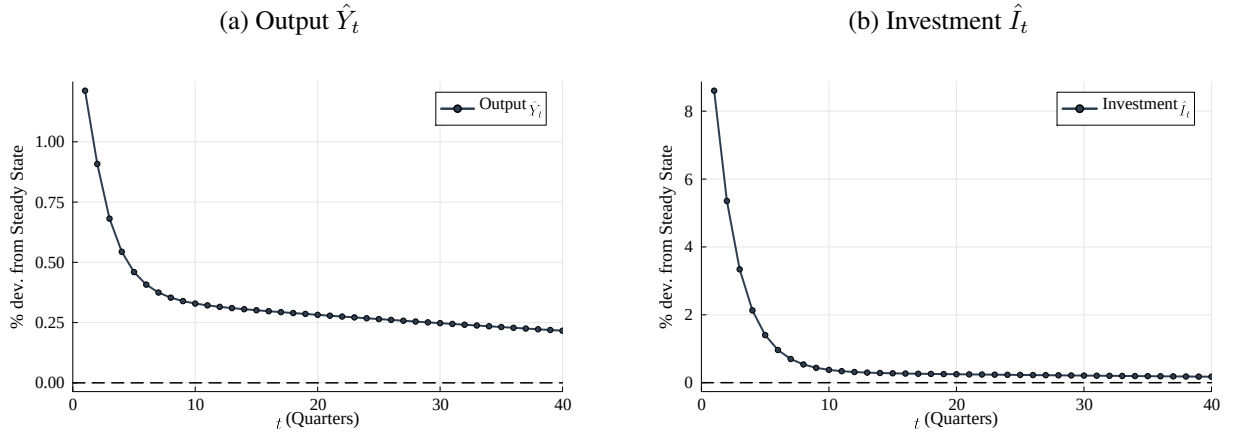


Figure 11: Impulse Responses to 100bp monetary shock

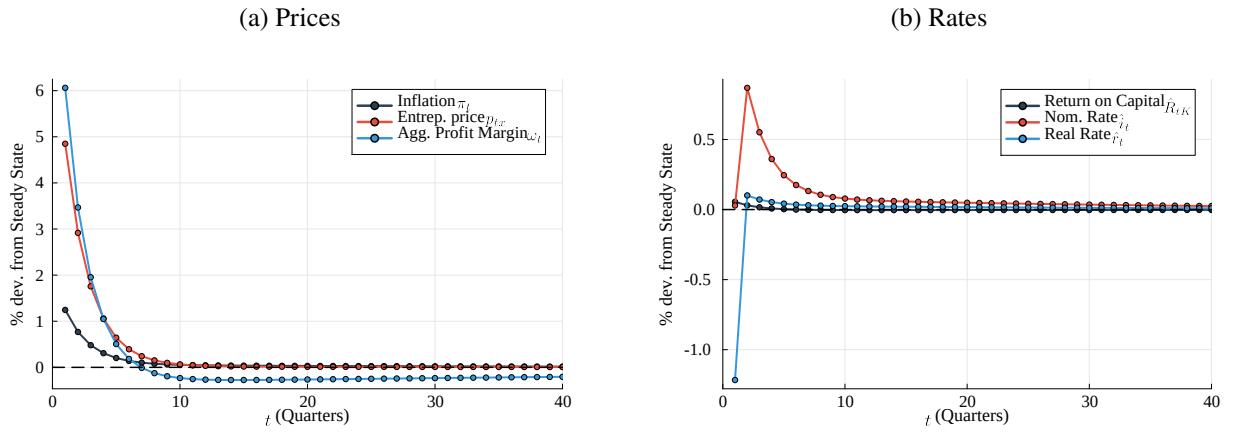


Figure 12: Amplification through Redistribution

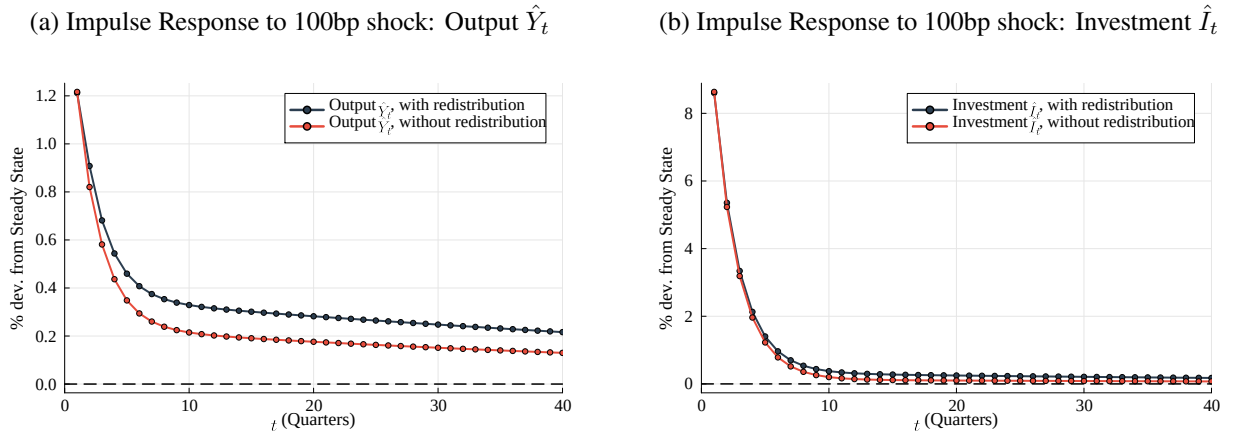


Figure 13: Phillips Curve Slope

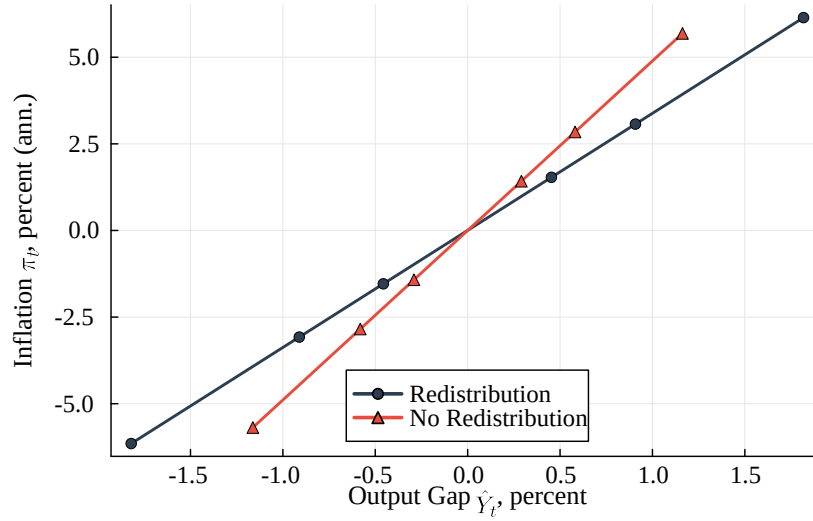


Figure 13 shows the flattening of the Phillips curve described theoretically in Section 3. For this experiment, I consider a range of monetary shocks ν_0 ranging from -200bp to 200bp. For each, I calculate the response of output $\hat{Y}_1$ and inflation π_1 in response to the shock, both in the period *after* the shock in order to account for any potential redistributive effects. I plot the former against the latter in Figure 13. I repeat this experiment for the two economies described immediately above: one with redistribution, and the other without, both starting from the same initial conditions. As demonstrated by Figure 13, allowing for changes in productivity resulting from redistribution meaningfully alters this relationship.

For the theoretical reasons discussed in Section 3, the rise in productivity implies that the marginal cost to producing an additional unit of output falls, which implies less inflation for a given increase in output. Figure 13 shows this directly: the slope of the output-inflation line is meaningfully smaller than in the economy without redistribution. Quantitatively, allowing for redistribution flattens the slope of the Phillips curve by about 45%. This is a relevant comparison for two reasons, one normative and one positive. On the positive side, standard HANK models assume that production is carried out by a representative producer, thereby shutting down the redistributive, supply-side channel in my model—analogous to the red line in Figure 13. Furthermore, when evaluating *optimal* monetary policy, the tradeoff between output and inflation is the fundamental choice faced by the policymaker. As a result, the optimal response of monetary policy to a given shock will be altered in the presence of redistribution.

5.4 INCREASE IN INEQUALITY

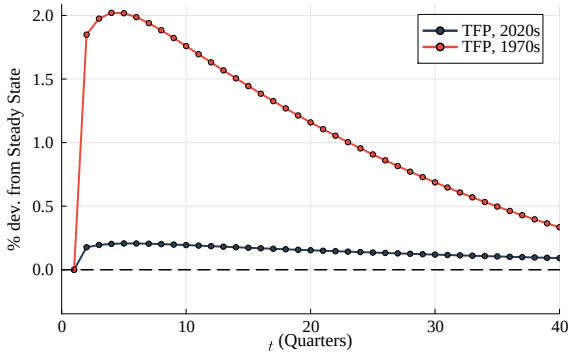
Since the early 1970s, two trends are present in the data. First, and famously, wealth inequality has increased: I measure using historical SCF data that the wealth Gini coefficient for the US has increased from 0.7 in 1971, to 0.85 in 2022. Concurrently, the efficacy of monetary policy shocks has decreased; for example, Boivin et al. (2010) estimate that the effect of an identified monetary shock on output is about half as large now as in

Table 2: Calibration, 1970s and 2010s

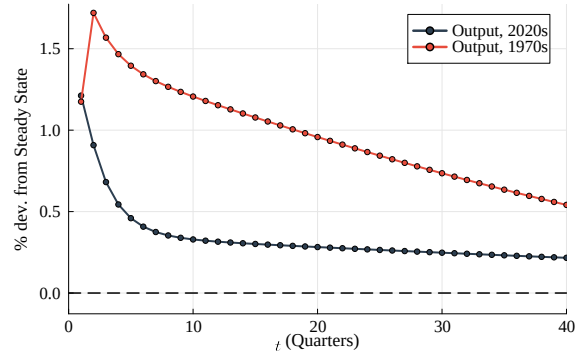
Parameter	Value, 1970s	Value, 2010s
p	0.980	0.985
σ_z	1.15	3.90
Gini	0.7	0.82

Figure 14: Impulse Responses: 1970 and 2022

(a) Productivity (TFP) Impulse Responses, 1970 and 2022



(b) Output Impulse Responses, 1970 and 2022



the 1960s and early 1970s. Because my model suggests that increases in inequality can dampen the effects of monetary policy, it offers a natural lens with which to explore whether these two trends are related.

To study this question, I consider two calibrations of my model. Holding all other parameters fixed, I adjust return persistence p and volatility σ_z to target two levels of wealth inequality: the Gini Coefficient in 1971, and the Gini coefficient in 2012, both measured in the SCF. Table 2 summarizes the two calibrations. My model attributes the increase in wealth inequality between these two years to an increase in both persistence *and* volatility.

The impulse responses in the two economies, presented in Figure 14, are markedly different. As compared to the 2022 economy, a monetary shock to the 1970s economy generates a much larger increase in productivity, as seen in Panel 14a. Per the results in Section 4, this change in the productivity suggests that as inequality has increased since the 1970s, the *reallocation* channel has dominated, and the efficacy of policy has been muted as the productivity channel has shut down. Because the productivity response is smaller in the 2022 economy, so too is the response of output, as shown in Panel 14b. Thus, my calibration suggests that the increase in wealth inequality since the 1970s is indeed partially responsible for the decrease in the efficacy of policy over the same period. Additionally, one finding in the literature on the changing efficacy of monetary policy over time is that, relative to the 1960s-1970s, the response of output in the post-1970s period to monetary policy peaks at a lower level, but persists for longer. This is exactly the pattern of output in Panel 14b: following the shock, output in the 2022 economy with higher wealth inequality peaks at a lower level, but remains elevated for longer than in the 1970 economy.

6 CONCLUSION

This paper has studied how the aggregate effects of a monetary policy rate change depend on wealth inequality. Over the past fifty years, as wealth inequality has increased, the measured effect of monetary policy on aggregates has declined. Existing models of monetary policy with heterogeneous households have struggled to reconcile these two trends: in these environments – where household labor income and marginal propensities to consume are the key sources of heterogeneity – wealth inequality has tended to amplify policy, contradicting the data.

I argue that the existing literature has overlooked the fact that households are also heterogeneous in their marginal propensities to *invest*, and in the returns that they earn on their wealth. A key component of this fact is leverage: wealthy, high-return households can take on debt to capitalize on a spread between their private returns and the cost of borrowing. I construct a model of heterogeneous entrepreneurs that can match the joint distribution of wealth and returns in the data, as well as the observed levels of debt backing the investments of wealthy households.

In this environment, a monetary policy rate cut increases aggregate productivity by redistributing assets to high-return households: these households see their nominal debts devalued, and their profit margins increase. However, the strength of this redistributive channel is hump-shaped in the degree of wealth concentration present when the policy change takes place: this channel vanishes when inequality is very low or very high, and is strongest at intermediate values of inequality. This non-monotonicity is due to countervailing forces: higher wealth concentration implies that redistribution following a monetary easing is well-targeted towards households with persistently high returns, but a greater concentration of wealth also shrinks the magnitude of this redistribution.

Calibrating this model to the data, I show that it can account for the increases of productivity and wealth inequality that have been shown to accompany monetary easings in the data. Furthermore, the model reconciles the parallel trends in wealth inequality and the efficacy of monetary policy. From the view of this model, the increase in wealth concentration since the 1970s has made it more difficult for monetary easings to redistribute towards high-return households than in the past. The increase in wealth inequality has thus dampened the effect of monetary policy on productivity in recent years, resulting in smaller aggregate effects of policy rate changes as observed in the data.

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A APPENDIX

A.1 PROOF OF PROPOSITION 1

Proof. I begin with aggregate output. Given the optimal choice of labor in Lemma 1, the output of an entrepreneur with productivity z and capital k_t is

$$y_t(z, k_t) = \frac{\omega_t}{\alpha} z k_t$$

Integrating over entrepreneurs using the stationary distribution $g_t(a, z)$, and using the fact that $k_t(a, z) = \lambda a$ if $z > \underline{z}_t$ and 0 otherwise, gives

$$\begin{aligned} Y_t &= \frac{\omega_t}{\alpha} \lambda \int_{\underline{z}}^{\infty} \int_0^{\infty} a z g_t(a, z) \, da \, dz \\ &= \frac{\omega_t}{\alpha} \lambda K_t \int_{\underline{z}}^{\infty} z s_t(z) \, dz \\ &= \frac{\omega_t}{\alpha} \lambda K_t X_t \end{aligned}$$

where

$$X_t = \int_{\underline{z}}^{\infty} z s_t(z) \, dz$$

From the labor market, we have

$$\begin{aligned} N_t &= \left(\frac{\omega_t}{\alpha} \right)^{\frac{1}{1-\alpha}} \lambda \int_{\underline{z}}^{\infty} \int_0^{\infty} a z g_t(a, z) \, da \, dz \\ &= \left(\frac{\omega_t}{\alpha} \right)^{\frac{1}{1-\alpha}} \lambda K_t X_t \end{aligned}$$

which implies

$$\omega_t = \alpha \left(\frac{N_t}{\lambda K_t X_t} \right)^{1-\alpha}$$

so production is

$$\begin{aligned} Y_t &= \frac{\omega_t}{\alpha} \lambda K_t X_t \\ &= \left(\frac{N_t}{\lambda K_t X_t} \right)^{1-\alpha} \lambda K_t X_t \\ &= (\lambda X_t K_t)^{\alpha} N_t^{1-\alpha} \end{aligned}$$

In order to eliminate λ , note that capital market clearing requires

$$\begin{aligned}
K_t &= \int_0^\infty \int_0^\infty k_t(a, z) g_t(a, z) \, da \, dz \\
&= \int_{\underline{z}}^\infty \int_0^\infty \lambda a g_t(a, z) \, da \, dz \\
&\downarrow \\
1 &= \lambda \int_{\underline{z}}^\infty s_t(z) \, dz
\end{aligned}$$

and thus

$$\lambda = \frac{1}{\int_{\underline{z}}^\infty s_t(z) \, dz}$$

Replacing this into production gives

$$Y_t = (Z_t K_t)^\alpha N_t^{1-\alpha}$$

where

$$\begin{aligned}
Z_t &= \lambda X_t \\
&= \frac{\int_{\underline{z}}^\infty z s_t(z) \, dz}{\int_{\underline{z}}^\infty s_t(z) \, dz} \\
&= \mathbb{E}_{s_t} [z | z > \underline{z}]
\end{aligned}$$

Now, the law of motion for the aggregate capital stock is

$$\begin{aligned}
K_{t+1} &= \int \int a_{t+1}(a, z) g_t(a, z) \, da \, dz \\
&= \int \int \beta R_t(z) a g_t(a, z) \, da \, dz \\
&= \beta K_t \int R_t(z) s_t(z) \, dz
\end{aligned}$$

Recall that

$$R(z_t) = 1 + r_t + \lambda \max \{ \omega_t z_t - r_t - \delta, 0 \}$$

and so I can write this as

$$\begin{aligned}
K_{t+1} &= \beta K_t \int [1 + r_t + \lambda \max \{ \omega_t z_t - r_t - \delta, 0 \}] s_t(z) \, dz \\
&= \beta K_t \left(1 + r_t + \lambda \int_{\underline{z}}^\infty (\omega_t z - r_t - \delta) s_t(z) \, dz \right) \\
&= \beta K_t \left(1 + r_t + \lambda \int_{\underline{z}}^\infty z \omega_t s_t(z) \, dz - \lambda (r_t + \delta) \int_{\underline{z}}^\infty s_t(z) \, dz \right)
\end{aligned}$$

The two integrals are

$$\begin{aligned}\omega_t \lambda \int_{\underline{z}}^{\infty} z s_t(z) \, dz &= \omega_t X_t \lambda \\ &= \omega_t Z_t\end{aligned}$$

and

$$\begin{aligned}\lambda \int_{\underline{z}}^{\infty} s_t \, dz &= \lambda (1 - S(\underline{z})) \\ &= 1\end{aligned}$$

and so the LoM is

$$K_{t+1} = \beta K_t (1 + \omega_t Z_t - \delta)$$

From labor market clearing,

$$\omega_t Z_t = \alpha p_{tx} Z_t^\alpha \left(\frac{N_t}{K_t} \right)^{1-\alpha}$$

and so the LoM becomes

$$\begin{aligned}K_{t+1} &= \alpha \beta p_{tx} (Z_t K_t)^\alpha N_t^{1-\alpha} + \beta (1 - \delta) K_t \\ &= \alpha \beta p_{tx} Y_t + \beta (1 - \delta) K_t\end{aligned}$$

The return on capital equals the average return across all entrepreneurs, calculated above:

$$\begin{aligned}R_{tK} &= \int_0^{\bar{z}} R_t(z) s_t(z) \, dz \\ &= 1 - \delta + \omega_t Z_t \\ &= 1 - \delta + \alpha p_{tx} Z_t^\alpha \left(\frac{N_t}{K_t} \right)^{1-\alpha} \, dz\end{aligned}$$

Finally, the factor prices. The wage can be calculated from labor market clearing: recall that

$$N_t = \left(\frac{1 - \alpha}{w_t} p_{tx} \right)^{\frac{1}{\alpha}} \lambda K_t X_t$$

Rearranging gives

$$w_t = (1 - \alpha) p_{tx} \left(\frac{Z_t K_t}{N_t} \right)^\alpha$$

as in the text. The net real interest rate comes from the definition of the cutoff $\underline{z}_t$:

$$r_t = \omega_t \underline{z}_t - \delta$$

Substituting the definition of ω_t from labor market clearing gives the form in Equation (28). Note that this only holds in expectation, as nominal debt contracts are negotiated at the end of period t , and the ex-post real return r_t depends on the realization of inflation. $\square$

A.2 PROOF OF PROPOSITION 2

Proof. By definition, the law of motion for the *cumulative* wealth share $S_t(z)$ is

$$S_{t+1}(z) = \frac{1}{K_{t+1}} p \int_0^z \int_0^\infty a'(a, \hat{z}) g_t(a, \hat{z}) da d\hat{z} + \frac{1}{K_{t+1}} (1-p) F(z) \int_0^\infty \int_0^\infty a'(a, \hat{z}) g_t(a, \hat{z}) da d\hat{z}$$

Differentiating with respect to z gives

$$S'_{t+1}(z) = \frac{1}{K_{t+1}} p \int_0^\infty a'(a, z) g_t(a, z) da + \frac{1}{K_{t+1}} (1-p) f(z) \int_0^\infty \int_0^\infty a'(a, \hat{z}) g_t(a, \hat{z}) da d\hat{z}$$

With the policy functions:

$$\begin{aligned} s_{t+1}(z) &= \frac{1}{K_{t+1}} p \int_0^\infty \beta R(z) a g_t(a, z) da + \\ &\quad \frac{1}{K_{t+1}} (1-p) f(z) \int_0^\infty \int_0^\infty \beta R(\hat{z}) a g_t(a, \hat{z}) da d\hat{z} \\ &= \frac{K_t}{K_{t+1}} p \beta R(z) s_t(z) + \\ &\quad \frac{K_t}{K_{t+1}} (1-p) f(z) \int_0^\infty \beta R(\hat{z}) s_t(\hat{z}) d\hat{z} \end{aligned}$$

Leibniz rule implies:

$$\frac{d}{dz} \int_0^z \int_0^\infty \beta R(\hat{z}) a g_t(a, \hat{z}) da d\hat{z} = \int_0^\infty \beta R(z) a g_t(a, z) da$$

Furthermore,

$$\int R(z) s_t(z) dz = 1 - \delta + \alpha p_{tx} Z_t^\alpha \left(\frac{N_t}{K_t} \right)^{1-\alpha}$$

as derived above, i.e. the RoK. So the LoM is

$$s_{t+1}(z) = \frac{K_t}{K_{t+1}} \beta \left[p R(z) s_t(z) + (1-p) f(z) \left(1 - \delta + \alpha p_{tx} Z_t^\alpha \left(\frac{N_t}{K_t} \right)^{1-\alpha} \right) \right] \quad (65)$$

Recall that from the law of motion for capital,

$$\frac{K_{t+1}}{\beta K_t} = 1 - \delta + \alpha p_{tx} Z_t^\alpha \left(\frac{N_t}{K_t} \right)^{1-\alpha} = R_{tK}$$

Substituting this into (65) gives Equation (33) in the text. $\square$

A.3 PROOF OF PROPOSITION 3

Proof. Starting from the log-linearized system in Section A.7, evaluating at $t = 0$ and combining gives $\hat{\omega}_0 = \xi \pi_0$ where

$$\xi = \frac{1 + \eta}{\alpha + \eta} \frac{1}{\kappa_p}$$

Similarly, the linearized response of the wealth shares $s_t(z)$ follows

$$\hat{s}_1(z) = \begin{cases} p\beta R(z) \pi_0 \left\{ \frac{\lambda-1}{R(z)} + \xi \left(\frac{\lambda\omega z}{R(z)} - \frac{\omega Z}{R_K} \right) \right\} & z > \underline{z} \\ p\beta R \pi_0 \left\{ -\frac{1}{R} - \xi \frac{\omega Z}{R_K} \right\} & z \leq \underline{z} \end{cases}$$

A.3.1 PROOF OF LEMMA 4

To simplify notation, let

$$q \equiv 1 - \beta(1 - \delta) = \frac{\omega Z}{R_K}$$

and

$$\begin{aligned} \phi(z) &= \lambda\omega z - qR(z) \\ &= \lambda\omega\beta(1 - \delta)z - q[(1 - \delta) - (\lambda - 1)\omega\underline{z}] \end{aligned}$$

With these, I can write

$$\hat{s}_1(z) = \begin{cases} p\beta\pi_0 \{\lambda - 1 + \xi\phi(z)\} & z > \underline{z} \\ p\beta\pi_0 \{-1 - \xi qR\} & z \leq \underline{z} \end{cases}$$

Clearly, $\hat{s}_1(z) < 0$ for $z \leq \underline{z}$. For $z > \underline{z}$, the sign shifts where $\phi(z) = -\frac{\lambda-1}{\xi}$, implying (after manipulation)

$$z^\dagger = \frac{1}{\lambda} \left[Z - \frac{q(\lambda - 1)}{1 - q} \underline{z} - \frac{\lambda - 1}{\beta\omega\xi(1 - \delta)} \right]$$

Finally, $z^\dagger < \underline{z}$ if

$$\underline{z} < \frac{1 - q}{\lambda - q} Z - \frac{\lambda - 1}{(\lambda - q)\omega\xi}$$

Multiplying through by $\omega > 0$ shows that the above inequality holds iff

$$r + \delta < \frac{(1 - \delta)q\xi - (\lambda - 1)}{(\lambda - q)\xi}$$

This is the inequality in Equation (43). □

A.4 PROOF OF PROPOSITION 4

Proof. A first-order approximation around the zero-inflation steady state gives

$$\hat{Z}_1 = \lambda \int_{\underline{z}}^{\bar{z}} (z - \underline{z}) \hat{s}_1(z) s(z) dz$$

Using the notation introduced in the proof of Proposition 3 in Section A.3, I can write

$$\begin{aligned} \hat{Z}_1 &= p\beta\pi_0(\lambda - 1) \int_{\underline{z}}^{\bar{z}} (z - \underline{z}) s(z) dz \\ &\quad + p\beta\pi_0\xi \int_{\underline{z}}^{\bar{z}} (z - \underline{z}) \phi(z) dz \end{aligned}$$

The term in the first line is positive and, as shown in the text of Proposition 4, proportional to aggregate debt to assets D/K in the economy. As such, it remains to show that the term on the second line – the contribution of the profit channel – is positive.

By Chebyshev's integral inequality,

$$\frac{1}{\lambda} \int_{\underline{z}}^{\bar{z}} (z - \underline{z}) \phi(z) s(z) dz \geq \left(\int_{\underline{z}}^{\bar{z}} (z - \underline{z}) s(z) dz \right) \left(\int_{\underline{z}}^{\bar{z}} \phi(z) s(z) dz \right)$$

The first integral on the right-hand side is clearly positive. The second evaluates to

$$\begin{aligned} \int_{\underline{z}}^{\bar{z}} \left[\lambda\omega z - \frac{\omega Z}{R_K} R(z) \right] s(z) dz &= \omega Z - [1 - \beta(1 - \delta)] \left(\frac{1}{\lambda} [1 - \delta - (\lambda - 1)\omega\underline{z}] + \omega Z \right) \\ &= \omega \frac{\lambda - 1}{\lambda} (\beta(1 - \delta)Z + [1 - \beta(1 - \delta)]\omega\underline{z}) > 0 \end{aligned}$$

Thus, both terms are positive, and $\hat{Z}_1 > 0$. □

A.5 PROOF OF PROPOSITION 5

The derivative of $\hat{s}_1(z)$ for $z > z^\dagger$ can be written as the sum of two terms:

$$\frac{d}{dp} \hat{s}_1(z) = \underbrace{\beta R(z) \{ \hat{R}_0(z) - \hat{R}_{0K} \}}_A + \underbrace{p\beta \frac{d}{dp} \left(R(z) \{ \hat{R}_0(z) - \hat{R}_{0K} \} \right)}_B$$

By definition, term $A = \hat{s}_1(z)/p > 0$ for $z > z^\dagger$. What remains to show is that term $B < 0$ when the condition in Equation (59) is satisfied. The proof proceeds as follows. To begin, I prove an intermediate lemma establishing how the cutoff $\underline{z}$ and aggregate profit margin ω respond as p changes. I then prove the main lemma, that $\partial_p \phi(z) < 0$.

Lemma 5. *In the long-run, zero-inflation steady state equilibrium defined in Section 2.5,*

$$\frac{d\underline{z}}{dp} > 0, \quad \text{and} \quad \frac{d\omega}{dp} < 0$$

Proof. Recall that the steady-state wealth shares satisfy

$$s(z) = \frac{1-p}{1-p\beta R(z)} f(z)$$

where returns are

$$R(z) = 1 - \delta + \omega [\lambda \max\{z - \underline{z}, 0\} + \underline{z}]$$

Next, note that the steady-state equilibrium can be written as a system of two equations in $(\underline{z}, \omega)$:

$$\int_0^{\underline{z}} s(z) dz = \frac{\lambda - 1}{\lambda} \tag{66}$$

$$\int_0^{\bar{z}} s(z) dz = 1 \tag{67}$$

The first ensures capital market clearing by fixing the mass of wealth *below* the cutoff $\underline{z}$. The second ensures that the shares integrate to one, which they do not generically.

Because returns below the cutoff are constant – $R(z) = 1 + \omega \underline{z} - \delta$ for $z \leq \underline{z}$, the first equation can be used to isolate ω as a function of $\underline{z}$:

$$\omega = \Omega(\underline{z}, p) = \frac{1}{\underline{z}} \left\{ \frac{1}{p\beta} \left[1 - \frac{(1-p)\lambda}{\lambda-1} F(\underline{z}) \right] - (1-\delta) \right\} \tag{68}$$

The derivatives of this object in its arguments are

$$\begin{aligned} \Omega_{\underline{z}} &= -\frac{1}{\underline{z}} \omega - \frac{1}{\underline{z}} \frac{1}{p\beta} \frac{(1-p)\lambda}{\lambda-1} f(\underline{z}) \\ \Omega_p &= \frac{1 - R\beta}{\underline{z}p(1-p)\beta} \end{aligned}$$

Unambiguously, $\Omega_p > 0$ and $\Omega_{\underline{z}} < 0$.

Next, note that combining equations (66) and (67) allows me to write

$$H(\underline{z}, p) = \lambda \int_{\underline{z}}^{\bar{z}} \frac{1-p}{1-p\beta R(z; \underline{z}, p)} f(z) dz - 1 \tag{69}$$

with $H(\underline{z}, p) = 0$ in equilibrium. Totally differentiating in p thus gives

$$\frac{d\underline{z}}{dp} = -\frac{H_p}{H_{\underline{z}}}$$

The derivatives of H are

$$H_p = \lambda \int_{\underline{z}}^{\bar{z}} \frac{\beta R(z) - 1 + p\beta(1-p)z_\lambda(z)\Omega_p}{1 - p\beta R(z)} \frac{s(z)}{1-p} dz$$

$$H_{\underline{z}} = \lambda \left(-s(\underline{z}) + p\beta \int_{\underline{z}}^{\bar{z}} \frac{\Omega_{\underline{z}} z_\lambda(z) - (\lambda-1)\omega}{1 - p\beta R(z)} s(z) dz \right)$$

Unambiguously $H_{\underline{z}} < 0$ (recall $\Omega_{\underline{z}} < 0$), so the sign of

$$\frac{d\underline{z}}{dp} = -\frac{H_p}{H_{\underline{z}}}$$

depends on the sign of H_p .

H_p can be decomposed as follows:

$$H_p = \lambda \int_{\underline{z}}^{\bar{z}} \frac{\beta R(z) - 1 + \beta p(1-p)z_\lambda(z)\Omega_p}{1 - p\beta R(z)} \frac{s(z)}{1-p} dz$$

$$= \lambda \left(\frac{1}{1-p} \int_{\underline{z}}^{\bar{z}} \frac{\beta R(z) - 1}{1 - p\beta R(z)} s(z) dz + \beta p \int_{\underline{z}}^{\bar{z}} \frac{z_\lambda(z)\Omega_p}{1 - p\beta R(z)} s(z) dz \right)$$

For a stationary wealth distribution to exist, we must have $p\beta R(z) < 1$ for all z , and so the second integral in parentheses is positive. For the first, note that I can write

$$\int_{\underline{z}}^{\bar{z}} \frac{\beta R(z) - 1}{1 - p\beta R(z)} s(z) dz = \underbrace{\int_0^{\bar{z}} \frac{\beta R(z) - 1}{1 - p\beta R(z)} s(z) dz}_{\mathcal{F}} - \underbrace{\int_0^{\underline{z}} \frac{\beta R(z) - 1}{1 - p\beta R(z)} s(z) dz}_{\mathcal{I}}$$

The integral $\mathcal{I}$ is

$$\mathcal{I} = \int_0^{\underline{z}} \frac{\beta R - 1}{1 - p\beta R} s(z) dz = \frac{\beta R - 1}{1 - p\beta R} \frac{\lambda - 1}{\lambda} < 0$$

For $\mathcal{F}$,

$$\mathcal{F} = \int_0^{\bar{z}} \frac{\beta R(z) - 1}{1 - p\beta R(z)} s(z) dz = \int_0^{\bar{z}} (\beta R(z) - 1) \frac{1}{1 - p\beta R(z)} s(z) dz$$

Both terms in the product under the integral are weakly increasing in z , and the first evaluates to

$$\int_0^{\bar{z}} (\beta R(z) - 1) s(z) dz = 0$$

As such, Chebyshev's integral inequality implies

$$\int_0^{\bar{z}} (\beta R(z) - 1) \frac{1}{1 - p\beta R(z)} s(z) dz \geq \left(\int_0^{\bar{z}} (\beta R(z) - 1) s(z) dz \right) \left(\int_0^{\bar{z}} \frac{1}{1 - p\beta R(z)} s(z) dz \right) = 0$$

and thus

$$\int_{\underline{z}}^{\bar{z}} \frac{\beta R(z) - 1}{1 - p\beta R(z)} s(z) dz = \mathcal{F} - \mathcal{I} > 0$$

As a result, $H_{\underline{z}} > 0$, and so $d\underline{z}/dp > 0$.

Turning to $d\omega/dp$, recall that in equilibrium $\omega = \Omega(\underline{z}, p)$. Totally differentiating gives

$$\frac{d\omega}{dp} = \Omega_{\underline{z}} \frac{d\underline{z}}{dp} + \Omega_p$$

Substituting $d\underline{z}/dp = -H_p/H_{\underline{z}}$, we have

$$\frac{d\omega}{dp} = \frac{\Omega_p H_{\underline{z}} - \Omega_{\underline{z}} H_p}{H_{\underline{z}}}$$

Recall that $H_{\underline{z}} < 0$ unambiguously, and so the sign of $d\omega/dp$ rests on $\Omega_p H_{\underline{z}} - \Omega_{\underline{z}} H_p$. This object is equal to

$$\begin{aligned} \Omega_p H_{\underline{z}} - \Omega_{\underline{z}} H_p &= -\Omega_p \lambda s(\underline{z}) - \Omega_p p \beta \lambda (\lambda - 1) \omega \zeta_1 \\ &\quad - \Omega_{\underline{z}} \frac{\lambda}{1-p} \zeta_3 \end{aligned}$$

where

$$\begin{aligned} \zeta_1 &\equiv \int_{\underline{z}}^{\bar{z}} \frac{1}{1-p\beta R(z)} s(z) dz \\ \zeta_3 &\equiv \int_{\underline{z}}^{\bar{z}} \frac{\beta R(z) - 1}{1-p\beta R(z)} s(z) dz \end{aligned}$$

Substituting in the appropriate derivatives for Ω and H , the term is positive if

$$\left[\frac{1}{\underline{z}} \omega + \frac{1}{\underline{z}} \frac{1}{p\beta} \frac{(1-p)\lambda}{\lambda-1} f(\underline{z}) \right] \frac{\lambda}{1-p} \zeta_3 > \frac{1-R\beta}{\underline{z}p(1-p)\beta} [\lambda s(\underline{z}) + p\beta\lambda(\lambda-1)\omega\zeta_1] \quad (70)$$

Using

$$s(\underline{z}) = \frac{1-p}{1-p\beta R} f(\underline{z})$$

and combining, the inequality (70) can be written as

$$\frac{1}{p\beta} f(\underline{z}) \left[\frac{\lambda}{\lambda-1} \zeta_3 - \frac{1-\beta R}{1-p\beta R} \right] > \omega [(1-R\beta)(\lambda-1)\zeta_1 - \zeta_3] \quad (71)$$

For the left-hand side of (71), note that

$$\int_0^{\underline{z}} \frac{1-\beta R(z)}{1-p\beta R(z)} s(z) dz = \frac{1-\beta R}{1-p\beta R} \frac{\lambda-1}{\lambda}$$

and so the term in brackets is equal to

$$\begin{aligned}
\frac{\lambda}{\lambda-1}\zeta_3 - \frac{1-\beta R}{1-p\beta R} &= \frac{\lambda}{\lambda-1} \int_{\underline{z}}^{\bar{z}} \frac{\beta R(z)-1}{1-p\beta R(z)} s(z) dz - \frac{\lambda}{\lambda-1} \int_0^{\underline{z}} \frac{1-\beta R}{1-p\beta R} s(z) dz \\
&= \frac{\lambda}{\lambda-1} \left[\int_{\underline{z}}^{\bar{z}} \frac{\beta R(z)-1}{1-p\beta R(z)} s(z) dz + \int_0^{\underline{z}} \frac{\beta R(z)-1}{1-p\beta R(z)} s(z) dz \right] \\
&= \frac{\lambda}{\lambda-1} \int_0^{\bar{z}} \frac{\beta R(z)-1}{1-p\beta R(z)} s(z) dz
\end{aligned}$$

By Chebyshev's inequality, the final line is positive, and thus the left-hand side of (71) is positive.

On the right-hand side of the inequality (71), by definition

$$(1-R\beta)(\lambda-1)\zeta_1 - \zeta_3 = \int_{\underline{z}}^{\bar{z}} \frac{(1-R\beta)(\lambda-1)}{1-p\beta R(z)} s(z) dz - \int_{\underline{z}}^{\bar{z}} \frac{\beta R(z)-1}{1-p\beta R(z)} s(z) dz$$

Therefore, the right-hand side of (71) is negative if

$$0 < \int_{\underline{z}}^{\bar{z}} \frac{\beta [R(z)-R] + \lambda(R\beta-1)}{1-p\beta R(z)} dz$$

Rearranging, and using the fact that $R(z)-R = \lambda\omega(z-\underline{z})$ and $1/\beta - R = \omega(Z-\underline{z})$, this equality can be shown to hold if

$$\begin{aligned}
Z - \underline{z} &< \int_{\underline{z}}^{\bar{z}} \frac{z - \underline{z}}{1-p\beta R(z)} \frac{s(z)}{\zeta_1} dz \\
&\updownarrow \\
\frac{1}{\lambda} \int_{\underline{z}}^{\bar{z}} (z - \underline{z}) s(z) dz &< \int_{\underline{z}}^{\bar{z}} (z - \underline{z}) \frac{s(z)}{1-p\beta R(z) \zeta_1} dz
\end{aligned} \tag{72}$$

Note that $s(z)/\lambda$ and

$$\frac{s(z)}{1-p\beta R(z)} \cdot \frac{1}{\zeta_1}$$

are both probability measures on $[\underline{z}, \bar{z}]$: both are everywhere nonnegative, and both integrate to one by construction on this interval. Furthermore, $1/[1-p\beta R(z)]$ is increasing in z on this interval. As such, the second measure places more mass on higher z values than the first, and the inequality in (72) strictly holds. Intuitively, both sides of (72) integrate the same increasing function of z , with the right-hand side using a measure that shifts weight to higher- z households relative to the measure on the left-hand side.

As a result, the inequality in (71) is guaranteed to hold: the left-hand side is positive, and the right-hand side negative. Therefore, $\Omega_p H_{\underline{z}} - \Omega_{\underline{z}} H_p > 0$, and

$$\frac{d\omega}{dp} < 0$$

□

With these results in hand, we can determine a condition under which Term B in $\partial_p \hat{s}_1(z)$ is negative.

Proof. Term B can be written as $p\beta\pi_0\xi \left\{ \frac{d}{dp} (\lambda\omega\bar{z} - qR(\bar{z})) \right\}$, where the term in braces is

$$\phi(z) = \lambda\omega\bar{z} - qR(z) = \lambda\beta(1-\delta)z\omega + q(\lambda-1)\omega\underline{z} - q(1-\delta)$$

The sign of term B, then, is determined by the sign of $\phi_p(z)$. This derivative is given by

$$\phi_p(z) = \lambda(1-q)z\frac{d\omega}{dp} + q(\lambda-1)\frac{d}{dp}(\omega\underline{z})$$

This derivative is negative when

$$\phi_p(\bar{z}) = \omega_p[\lambda(1-q)\bar{z} + q(\lambda-1)\underline{z}] + q(\lambda-1)\omega\underline{z}_p < 0$$

or equivalently, when

$$\varepsilon_{\omega,p}[\lambda(1-q)\bar{z} + q(\lambda-1)\underline{z}] + q(\lambda-1)\underline{z}\varepsilon_{\underline{z},p} < 0$$

where

$$\varepsilon_{\underline{z},p} = \frac{p}{\underline{z}}\underline{z}_p \quad \varepsilon_{\omega,p} = \frac{p}{\omega}\omega_p \quad (73)$$

are the elasticities of the cutoff $\underline{z}$ and return ω in p . Note that per Lemma 5, $\varepsilon_{\underline{z},p} > 0$ and $\varepsilon_{\omega,p} < 0$.

Rearranging equation (73) shows that $\phi_p(z) < 0$ when

$$|\varepsilon_{\omega,p}| > \frac{q(\lambda-1)\underline{z}}{\lambda(1-q)z + q(\lambda-1)\underline{z}}\varepsilon_{\underline{z},p}$$

which is the condition in equation (59) in the text. $\square$

A.6 PROOF OF PROPOSITION 6

Recall from Proposition 6 that one can write the derivative of $\hat{Z}_1$ – the response of aggregate productivity to a fixed price shock π_0 – as the sum of three terms:

$$\begin{aligned} \frac{d}{dp}\hat{Z}_1 &= -\left(\frac{d\underline{z}}{dp}\right)\mathbb{E}_s[\hat{s}_1(z) \mid z > \underline{z}] && \text{1. Participation} \\ &+ \mathbb{E}_s\left[(z - \underline{z}) \times \frac{d\hat{s}_1(z)}{dp} \mid z > \underline{z}\right] && \text{2. Redistribution} \\ &+ \mathbb{E}_s\left[(z - \underline{z})\hat{s}_1(z) \times \left(\frac{1}{s(z)}\frac{ds(z)}{dp}\right) \mid z > \underline{z}\right] && \text{3. Reweighting} \end{aligned}$$

Proof. Begin with Term 1: Participation. Note that this term is equal to $-\underline{z}_p\hat{s}_1(Z)$: the derivative of the cutoff $\underline{z}$ in p , multiplied by the wealth share change for the investor whose idiosyncratic productivity coincides with aggregate productivity Z . This investor has productivity which is average *among active investors* $z > \underline{z}$.

By Lemma 4, $\hat{s}_1(Z)$ is positive if $z^\dagger < Z$, or

$$\frac{1}{\lambda} \left[Z - \frac{q(\lambda - 1)}{1 - q} \underline{z} - \frac{\lambda - 1}{\beta \omega \xi (1 - \delta)} \right] < Z$$

Rearranging shows that this condition is equivalent to

$$-\frac{q}{1 - q} \underline{z} - \frac{1}{\beta \omega \xi (1 - \delta)} < Z$$

which is always the case. As such, Term 1 is unambiguously negative, as $\underline{z}_p > 0$ by Lemma 5.

Now, turn to Term 2: Redistribution. This term is equal to

$$\lambda \int_{\underline{z}}^{\bar{z}} (z - \underline{z}) \left(\frac{d}{dp} \hat{s}_1(z) \right) s(z) dz$$

Note that for $z > \underline{z}$, the derivative of the wealth share change $\hat{s}_1(z)$ can be written as $\frac{d}{dp} \hat{s}_1(z) = \beta \pi_0 (Gz + H)$, where

$$\begin{aligned} G &= \xi \lambda (1 - q) \omega (1 + \varepsilon_{\omega, p}) \\ H &= \lambda - 1 - \xi q (1 - \delta) + \xi q (\lambda - 1) \omega \underline{z} (1 + \varepsilon_{\omega \underline{z}, p}) \end{aligned}$$

With this notation, Term 2 becomes

$$\begin{aligned} \lambda \int_{\underline{z}}^{\bar{z}} (z - \underline{z}) \left(\frac{d}{dp} \hat{s}_1(z) \right) s(z) dz &= \lambda \int_{\underline{z}}^{\bar{z}} (z - \underline{z}) \beta \pi_0 (Gz + H) s(z) dz \\ &= \lambda \beta \pi_0 G \int_{\underline{z}}^{\bar{z}} (z - \underline{z}) z s(z) dz + \lambda \beta \pi_0 H \int_{\underline{z}}^{\bar{z}} (z - \underline{z}) s(z) dz \end{aligned}$$

The two integrals evaluate to

$$\begin{aligned} \lambda \int_{\underline{z}}^{\bar{z}} (z - \underline{z}) s(z) dz &= Z - \underline{z} \\ \lambda \int_{\underline{z}}^{\bar{z}} (z - \underline{z}) z s(z) dz &= \mathbb{E}_s [z^2 | z \geq \underline{z}] - \underline{z} Z \end{aligned}$$

Letting $V \equiv \text{Var}_s [z | z \geq \underline{z}]$ denote the wealth-weighted variance of productivities among active investors, the second integral becomes

$$\begin{aligned} \lambda \int_{\underline{z}}^{\bar{z}} (z - \underline{z}) z s(z) dz &= \mathbb{E}_s [z^2 | z \geq \underline{z}] - \underline{z} Z \\ &= V + Z (Z - \underline{z}) \end{aligned}$$

which implies that Term 2 can be written as

$$\begin{aligned} \lambda \beta \pi_0 G \int_{\underline{z}}^{\bar{z}} (z - \underline{z}) z s(z) dz + \lambda \beta \pi_0 H \int_{\underline{z}}^{\bar{z}} (z - \underline{z}) s(z) dz &= \beta \pi_0 G [V + Z (Z - \underline{z})] + \beta \pi_0 H (Z - \underline{z}) \\ &= \beta \pi_0 [GV + (Z - \underline{z}) (GZ + H)] \end{aligned} \quad (74)$$

This is positive wherever $GV + (Z - \underline{z}) (GZ + H) > 0$. To establish conditions under which this inequality holds, note that

$$\begin{aligned} GZ + H &= \frac{1}{\beta \pi_0} \frac{d}{dp} \hat{s}_1(Z) \\ &= \frac{\hat{s}_1(Z)}{p \beta \pi_0} + \xi \omega [\lambda (1 - q) Z \varepsilon_{\omega, p} + q (\lambda - 1) \underline{z} \varepsilon_{\omega \underline{z}, p}] \end{aligned}$$

Dividing through by $\xi \lambda (1 - q) \omega > 0$ gives

$$\frac{GZ + H}{\xi \lambda (1 - q) \omega} = \frac{\hat{s}_1(Z)}{p \beta \pi_0 \xi \lambda (1 - q) \omega} - Z |\varepsilon_{\omega, p}| + \frac{q (\lambda - 1)}{\lambda (1 - q)} \underline{z} \varepsilon_{\omega \underline{z}, p} \quad (75)$$

Let

$$\Delta(Z) \equiv \frac{\hat{s}_1(Z)}{p \beta \pi_0 \xi \lambda (1 - q) \omega} = \frac{\lambda - 1}{\lambda} \left[\frac{1}{\xi (1 - q) \omega} + Z + \frac{q}{1 - q} \underline{z} \right]$$

Dividing both sides of equation (74) by the same $\xi \lambda (1 - q) \omega$ factor gives

$$\begin{aligned} \frac{\lambda \int_{\underline{z}}^{\bar{z}} (z - \underline{z}) z \left(\frac{d}{dp} \hat{s}_1(z) \right) s(z) dz}{\beta \pi_0 \xi \lambda (1 - q) \omega} &= \frac{GV + (Z - \underline{z}) (GZ + H)}{\xi \lambda (1 - q) \omega} \\ &= V (1 - |\varepsilon_{\omega, p}|) - Z |\varepsilon_{\omega, p}| (Z - \underline{z}) \\ &\quad + (Z - \underline{z}) \left[\Delta(Z) + \frac{q (\lambda - 1)}{\lambda (1 - q)} \underline{z} \varepsilon_{\omega \underline{z}, p} \right] \end{aligned}$$

where the second equality follows from (75). Substituting $\varepsilon_{\omega \underline{z}, p} = \varepsilon_{\underline{z}, p} - |\varepsilon_{\omega, p}|$ and rearranging gives

$$\lambda \int_{\underline{z}}^{\bar{z}} (z - \underline{z}) z \left(\frac{d}{dp} \hat{s}_1(z) \right) s(z) dz \iff |\varepsilon_{\omega, p}| > \frac{V + (Z - \underline{z}) \left(\Delta(Z) + \frac{q (\lambda - 1)}{\lambda (1 - q)} \underline{z} \varepsilon_{\underline{z}, p} \right)}{V + (Z - \underline{z}) \left(Z + \frac{q (\lambda - 1)}{\lambda (1 - q)} \underline{z} \right)}$$

This is the condition (61) in Proposition 6 in the text.

Finally, Term 3: Reweighting is given by $\lambda \int_{\underline{z}}^{\bar{z}} (z - \underline{z}) \hat{s}_1(z) \left(\frac{d}{dp} s(z) \right) dz$. Let

$$\begin{aligned} w(z) &\equiv (z - \underline{z}) \hat{s}_1(z) \\ h(z) &\equiv \frac{d}{dp} s(z) \end{aligned}$$

so that

$$\lambda \int_{\underline{z}}^{\bar{z}} (z - \underline{z}) \hat{s}_1(z) \left(\frac{d}{dp} s(z) \right) dz = \lambda \int_{\underline{z}}^{\bar{z}} w(z) h(z) s(z) dz$$

As such, Term 3 is $\text{Cov}_s(w, h \mid z > \underline{z}) + \mathbb{E}_s[w \mid z > \underline{z}] \mathbb{E}_s[h \mid z > \underline{z}]$. Note that by construction, $\mathbb{E}_s[w \mid z > \underline{z}] = \hat{Z}_1$. Furthermore, in equilibrium $\lambda \int_{\underline{z}}^{\bar{z}} s(z) dz = 1$. Differentiating this identity in p gives $\mathbb{E}_s[h \mid z > \underline{z}] = \lambda s(\underline{z}) \underline{z}_p > 0$ by Lemma 5. As both conditional means are positive, so too is their product. As such, Term 3 is guaranteed to be positive wherever $\text{Cov}_s(w, h \mid z > \underline{z})$. Again by construction, w is increasing in z . As such, the covariance is positive as long as $h'(z) > 0$. Recall that

$$h(z) = \frac{d}{dp} s(z) = \frac{\beta R(z) - 1 + p(1-p)\beta [z_\lambda(z) \omega_p - (\lambda - 1) \omega_{\underline{z}p}]}{(1 - p\beta R(z))(1 - p)}$$

Substituting for $R(z)$ and $z_\lambda(z)$ gives

$$h(z) = \frac{Az + B}{Cz + D}$$

where

$$\begin{aligned} A &= \beta\lambda\omega(1 + (1-p)\varepsilon_{\omega,p}) & B &= -q - \beta(\lambda - 1)\omega\underline{z} [1 + (1-p)(\varepsilon_{\underline{z},p} + \varepsilon_{\omega,p})] \\ C &= -(1-p)p\beta\lambda\omega & D &= (1-p)[1 - p(1-q) + p\beta(\lambda - 1)\omega\underline{z}] \end{aligned}$$

which implies $h'(z) = (AD - BC) \times (Cz + D)^{-2}$, and the sign depends on the sign of $AD - BC$. Substituting the forms above and simplifying gives

$$\frac{1}{(1-p)^2 \beta\lambda\omega} (AD - BC) = 1 - |\varepsilon_{\omega,p}| [1 - p(1-q)] - p\beta(\lambda - 1)\omega\underline{z}\varepsilon_{\underline{z},p}$$

As such, we have the condition:

$$|\varepsilon_{\omega,p}| < \frac{1 - p\beta(\lambda - 1)\omega\underline{z}\varepsilon_{\underline{z},p}}{1 - p(1-q)} \implies \lambda \int_{\underline{z}}^{\bar{z}} (z - \underline{z}) \hat{s}_1(z) \left(\frac{d}{dp} s(z) \right) dz > 0$$

which is the condition in equation (62) in the text. □

A.7 FULL LOG-LINEARIZED SYSTEM

The full system is

$$\hat{s}_{t+1}(z) = p \frac{R(z)}{R_K} \left\{ \hat{R}_t(z) - \hat{R}_{tK} + \hat{s}_t(z) \right\} \quad (76)$$

$$\hat{R}_t(z) = \begin{cases} \frac{1}{R(z)} \left\{ (1-\lambda) (\hat{i}_t - \hat{\pi}_t) + \lambda \omega z \hat{\omega}_t \right\} & z > \underline{z}_t \\ \frac{1}{R} \left\{ \hat{i}_t - \hat{\pi}_t \right\} & z < \underline{z}_t \end{cases} \quad (77)$$

$$\hat{R}_{tK} = r_K \left\{ \hat{\omega}_t - \hat{Z}_t \right\} \quad (78)$$

$$\hat{\omega}_t = \hat{p}_{tx} + (1-\alpha) (\hat{N}_t - \hat{Z}_t - \hat{K}_t) \quad (79)$$

$$\hat{w}_t = \eta \hat{N}_t \quad (80)$$

$$\pi_t = \kappa_p \hat{p}_{tx} + \beta_f \mathbb{E}_t \pi_{t+1} \quad (81)$$

$$\hat{i}_t = \phi_\pi \pi_t + \nu_t \quad (82)$$

$$\hat{r}_t = \hat{i}_{t-1} - \pi_t \quad (83)$$

$$\hat{\underline{z}}_{t+1} = \frac{1}{r + \delta} \hat{r}_{t+1} - \hat{\omega}_{t+1} \quad (84)$$

$$\hat{N}_t = \frac{1}{\alpha + \eta} \hat{p}_{tx} + \frac{\alpha}{\alpha + \eta} (\hat{Z}_t + \hat{K}_t) \quad (85)$$

$$\hat{Y}_t = \alpha (\hat{Z}_t + \hat{K}_t) + (1-\alpha) \hat{N}_t \quad (86)$$

$$\hat{K}_{t+1} = [1 - \beta(1-\delta)] (\hat{p}_{tx} + \hat{Y}_t) + \beta(1-\delta) \hat{K}_t \quad (87)$$

$$\hat{Z}_{t+1} = \lambda \left\{ \underline{z}s(\underline{z}) \left(1 - \frac{\underline{z}}{Z} \right) \hat{\underline{z}}_{t+1} + \int_{\underline{z}}^{\bar{z}} \left(\frac{z}{Z} - 1 \right) s(z) \hat{s}_{t+1}(z) \, dz \right\} \quad (88)$$

$$0 = \underline{z}s(\underline{z}) \hat{\underline{z}}_t + \int_0^{\bar{z}} s(z) \hat{s}_t(z) \, dz \quad (89)$$

A.8 COMPUTATIONAL STRATEGY

A.8.1 STEADY STATE

The equations pinning down the steady state are as follows:

$$s(z) = \frac{1-p}{1-p\beta R(z)} f(z) \quad (90)$$

$$R(z) = \begin{cases} 1+r+\lambda(\omega z - r - \delta) & z > \underline{z} \\ 1+r & z \leq \underline{z} \end{cases} \quad (91)$$

$$\omega = \alpha p_x \left(\frac{N}{ZK} \right)^{1-\alpha} \quad (92)$$

$$p_x = \frac{\varepsilon - 1}{\varepsilon} \quad (93)$$

$$\frac{1}{\beta} = 1 + \alpha p_x Z^\alpha \left(\frac{N}{K} \right)^{1-\alpha} \quad (94)$$

$$N = [(1-\alpha)p_x]^{\frac{1}{\alpha+\eta}} (ZK)^{\frac{\alpha}{\alpha+\eta}} \quad (95)$$

$$\underline{z} = \frac{r + \delta}{\omega} \quad (96)$$

$$Z = \lambda \int_{\underline{z}}^{\bar{z}} z s(z) dz \quad (97)$$

$$1 = \lambda \int_{\underline{z}}^{\bar{z}} s(z) dz \quad (98)$$

My assumption on the process for z_t allows the computation of the steady state to be reduced to a system of two equations in two unknowns. I guess a pair (ω^0, r^0) . From there, I calculate

$$\begin{aligned} \underline{z}^0 &= \frac{r^0 + \delta}{\omega^0} \\ Z^0 &= \frac{1/\beta - (1-\delta)}{\omega^0} \end{aligned}$$

where the second follows from stationarity of the capital stock. Aggregate productivity then pins down the capital-labor ratio:

$$\begin{aligned} \frac{K^0}{N^0} &= \left(\frac{\alpha p_x (Z^0)^\alpha}{\frac{1}{\beta} - (1-\delta)} \right)^{\frac{1}{1-\alpha}} \\ &\equiv \xi^0 \end{aligned}$$

and then labor market clearing pins down equilibrium labor N^0 :

$$N^0 = [(1-\alpha)p_x (Z^0 \xi^0)^\alpha]^{1/\eta}$$

and then $K^0 = \xi^0 N^0$. Now, I check errors on the implied definitions of Z^0 and $\underline{z}^0$. I calculate $\underline{z}^1$ using capital market clearing: given r^0 and ω^0 , I can calculate the wealth shares $s^0(z)$, and then $\underline{z}^1$ solves

$$1 - \frac{1}{\lambda} = \int_{\underline{z}^1}^{\bar{z}} s^0(z) dz$$

and Z^1 is given by

$$Z^1 = \int_{\underline{z}^0}^{\bar{z}} z s^0(z) dz$$

I define a discrete grid for z on $[0, \bar{z}]$, and compute the above integrals using the Trapezoidal rule. As a starting guess, I set

$$r^0 = r^{FB} = \frac{1}{\beta}$$

and

$$\omega^0 = \omega^{FB} = \frac{1/\beta - (1 - \delta)}{\bar{z}}$$

equal to their values under the first-best equilibrium.

A.8.2 TRANSITIONS

I use the following algorithm to compute impulse responses to a monetary shock in my linearized model:

1. Start with $\hat{K}_0, \hat{s}_0(z), \hat{\underline{z}}_0, \hat{Z}_0, \hat{i}_0$ all equal to zero.
2. Guess a path for $\{\hat{i}_{t+1}\}_{t=0}^{T-1}$. For boundary conditions, we have $\hat{i}_0 = \hat{i}_{T+1} = 0$.
3. Calculate the path for π_t from the Taylor Rule:

$$\pi_t = \frac{\hat{i}_{t+1} - \nu_t}{\phi_t}$$

4. Using π_t and π_{t+1} , calculate $\hat{p}_{tx}$ from the Phillips curve:

$$\pi_t = \kappa_p \hat{p}_{tx} + \beta_f \mathbb{E}_t \pi_{t+1}$$

$$\hat{p}_{tx} = \frac{1}{\kappa_p} \{\pi_t - \beta \pi_{t+1}\}$$

5. Calculate $\hat{N}_t$:

$$\hat{N}_t = \frac{1}{\alpha + \eta} \hat{p}_{tx} + \frac{\alpha}{\alpha + \eta} (\hat{Z}_t + \hat{K}_t)$$

6. Calculate $\hat{\omega}_t, \hat{r}_t, \hat{R}_{tK}, \hat{R}_t(z)$:

$$\begin{aligned}\hat{R}_t(z) &= \begin{cases} \frac{1}{\hat{R}(z)} \left\{ (1-\lambda) (\hat{i}_t - \hat{\pi}_t) + \lambda \omega z \hat{\omega}_t \right\} & z > \underline{z}_t \\ \frac{1}{\hat{R}} \left\{ \hat{i}_t - \hat{\pi}_t \right\} & z < \underline{z}_t \end{cases} \\ \hat{R}_{tK} &= r_K \left\{ \hat{\omega}_t - \hat{Z}_t \right\} \\ \hat{\omega}_t &= \hat{p}_{tx} + (1-\alpha) (\hat{N}_t - \hat{Z}_t - \hat{K}_t) \\ \hat{r}_t &= \hat{i}_t - \pi_t\end{aligned}$$

- Note the notation in $\hat{R}_t(z)$: it's important that we not let them switch ex-post to investing after the initial unanticipated π_0 .

7. Using the returns, update the wealth shares:

$$\hat{s}_{t+1}(z) = p \frac{\hat{R}(z)}{\hat{R}_K} \left\{ \hat{R}_t(z) - \hat{R}_{tK} + \hat{s}_t(z) \right\}$$

8. Using the continuation shares $\hat{s}_{t+1}(z)$, find $\hat{\underline{z}}_{t+1}$ from capital market clearing:

$$\hat{\underline{z}}_{t+1} = \frac{1}{s(\underline{z})} \int_{\underline{z}}^{\bar{z}} \hat{s}_{t+1}(z) dz$$

9. Repeat steps 3-7 for all t . Then, evaluate the $\underline{z}_{t+1}$ values implied by capital market clearing against their definition: for $t = 0, \dots, T-1$:

$$\hat{\underline{z}}_{t+1} \stackrel{?}{=} \frac{1}{r + \delta} \hat{r}_{t+1} - \hat{\omega}_{t+1}$$

If these are all satisfied, stop. Otherwise, update the path $\{\hat{i}_t\}$ and return to step 2.